



# ASUS ZB555KL Maintenance Manual

# Technical documents

file name: **ASUS ZB555KL** Maintenance Manual

version : V1.0

Fake :  
Review :  
Will sign:

# ASUS ZB555KL Maintenance Manual

## directory

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**Required equipment and tools**

**ID appearance**

**Top view of the board**

**System Block Diagram**

**I2C address and power management**

**Important devices and key test points**

**Maintenance methods**



CMU200 OR 8960



DC power supply



scilloscope



Hot air gun



Electric iron



pectrum Analyzer



electric screw driver



UP-1024 High speed programmer



Disassembly tool



isassembly tool



multimeter



Anti-static gloves



Lead-free solder  
wire



Anti-static  
cloth



alcohol bottle



Pen brush



Anti-static brush



Anti-static fixture



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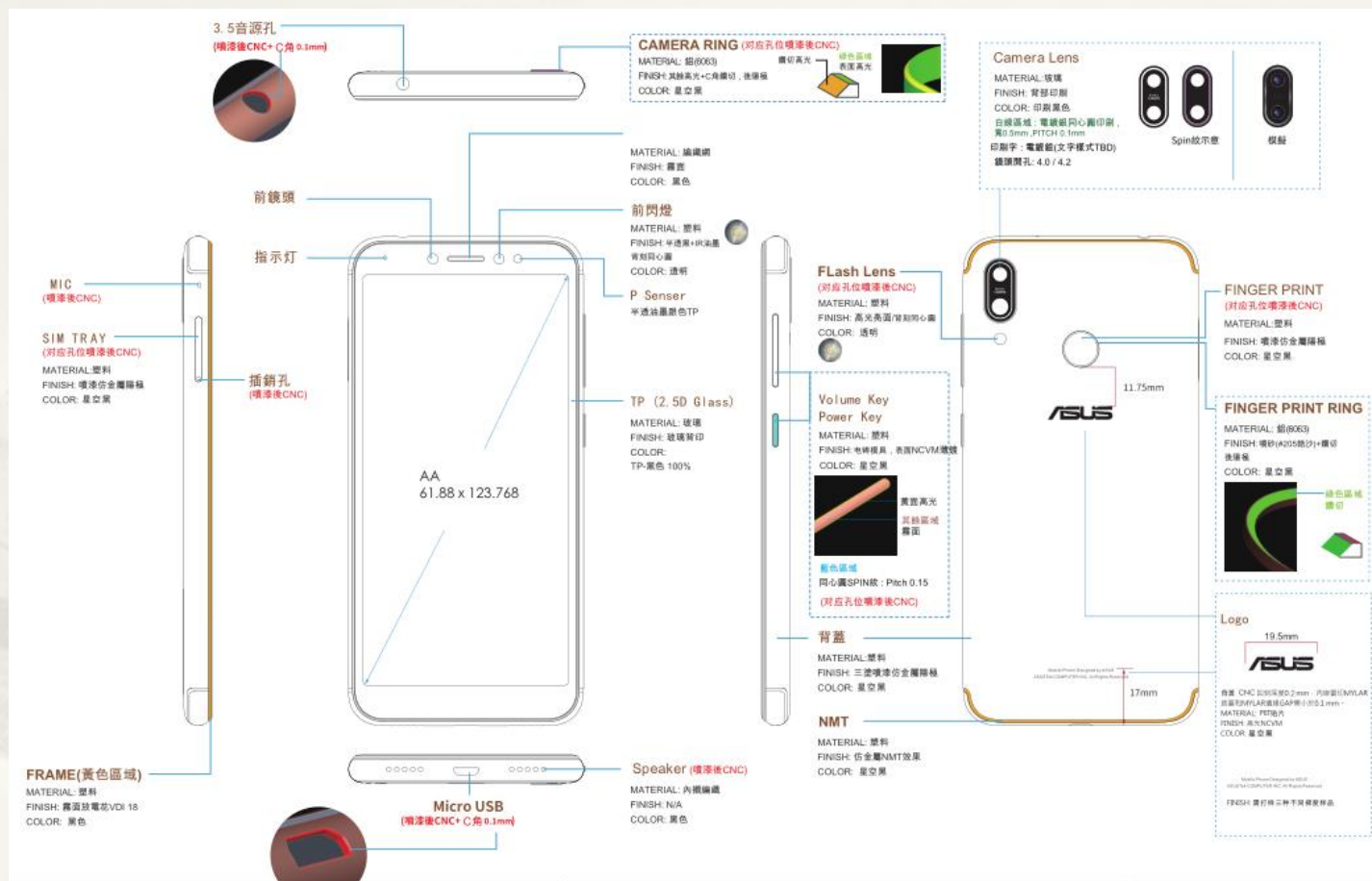
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# ID Exterior





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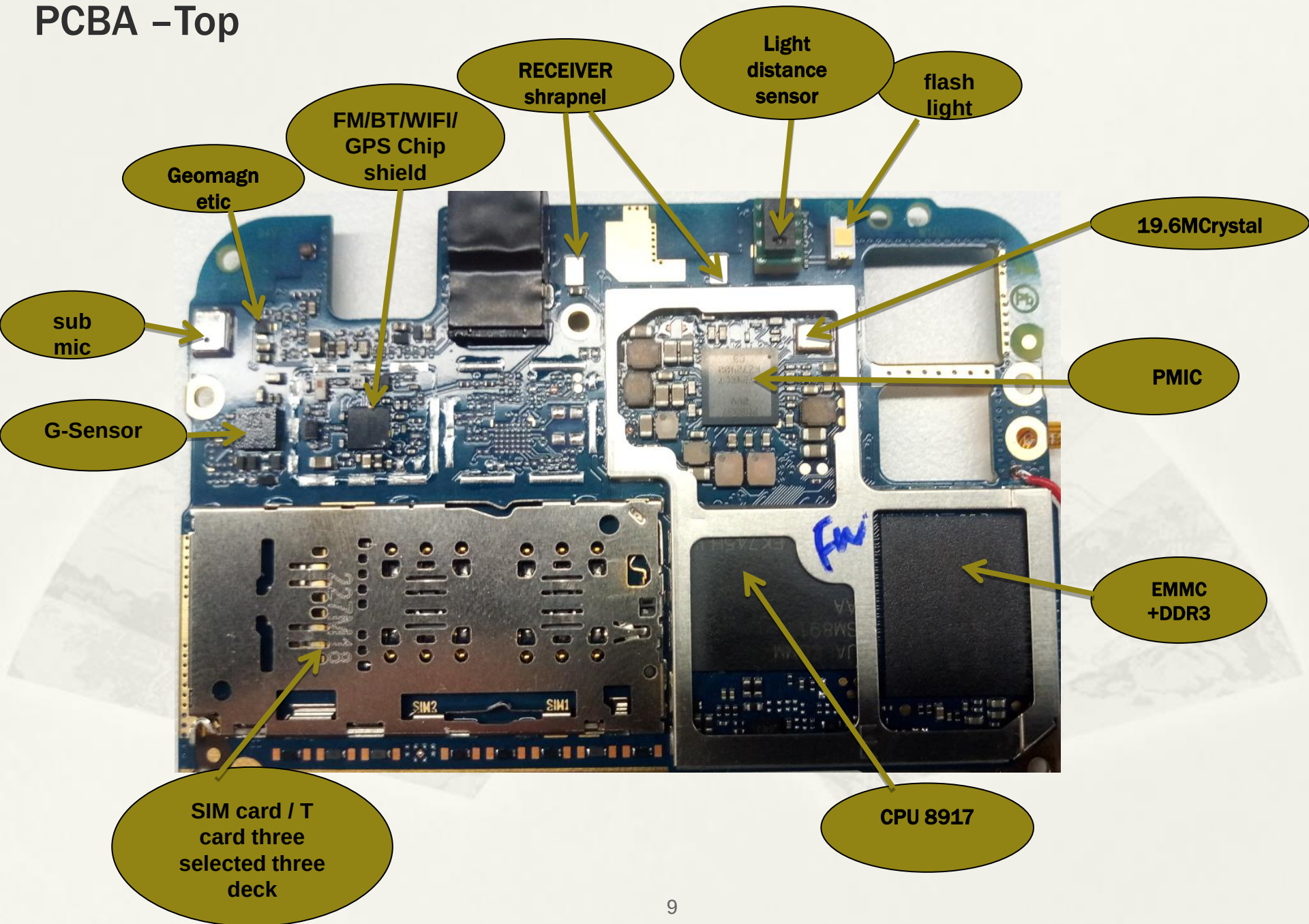
**System Block Diagram**

**I2C address and power management**

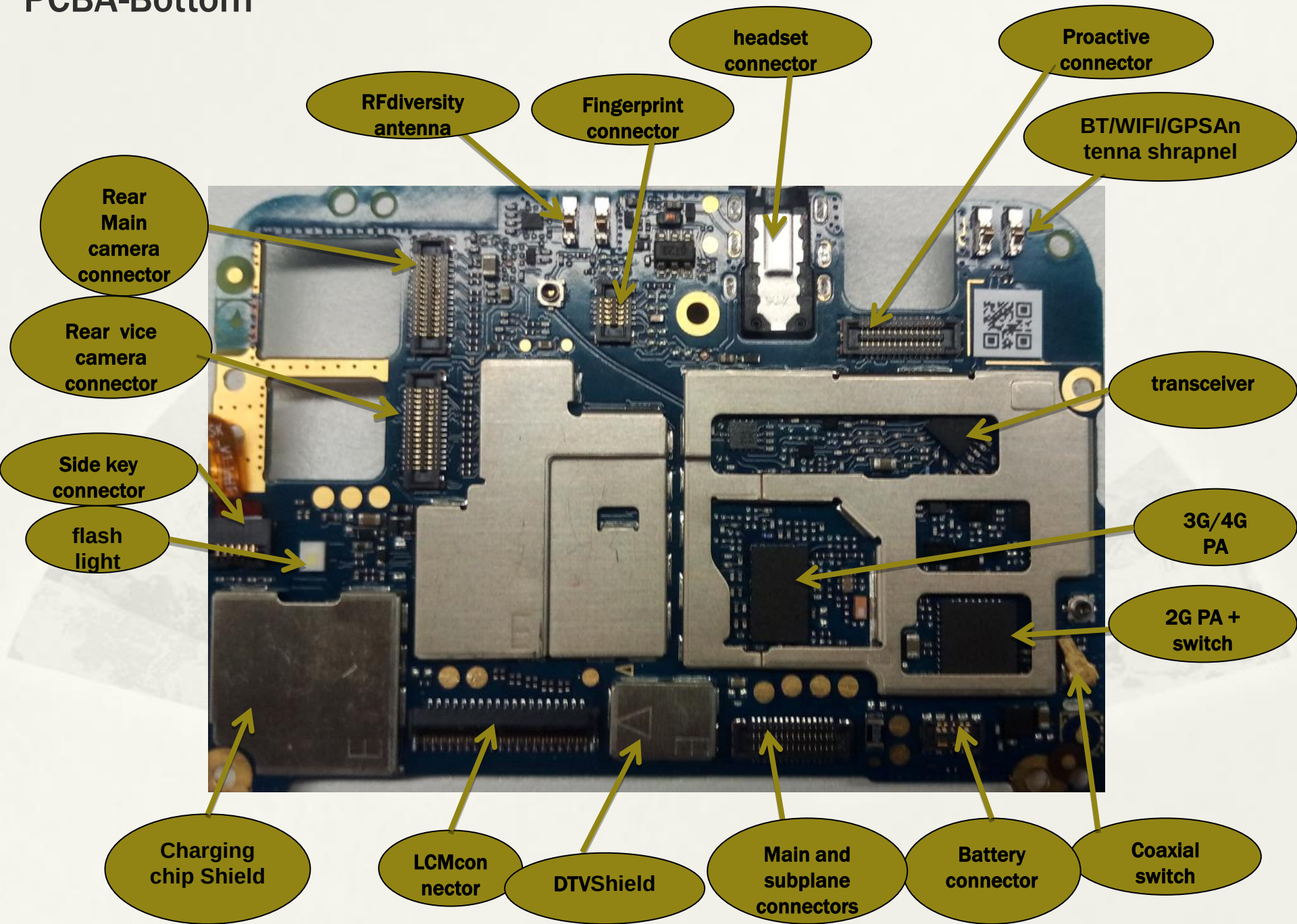
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# PCBA -Top



# PCBA-Bottom





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# System Block Diagram

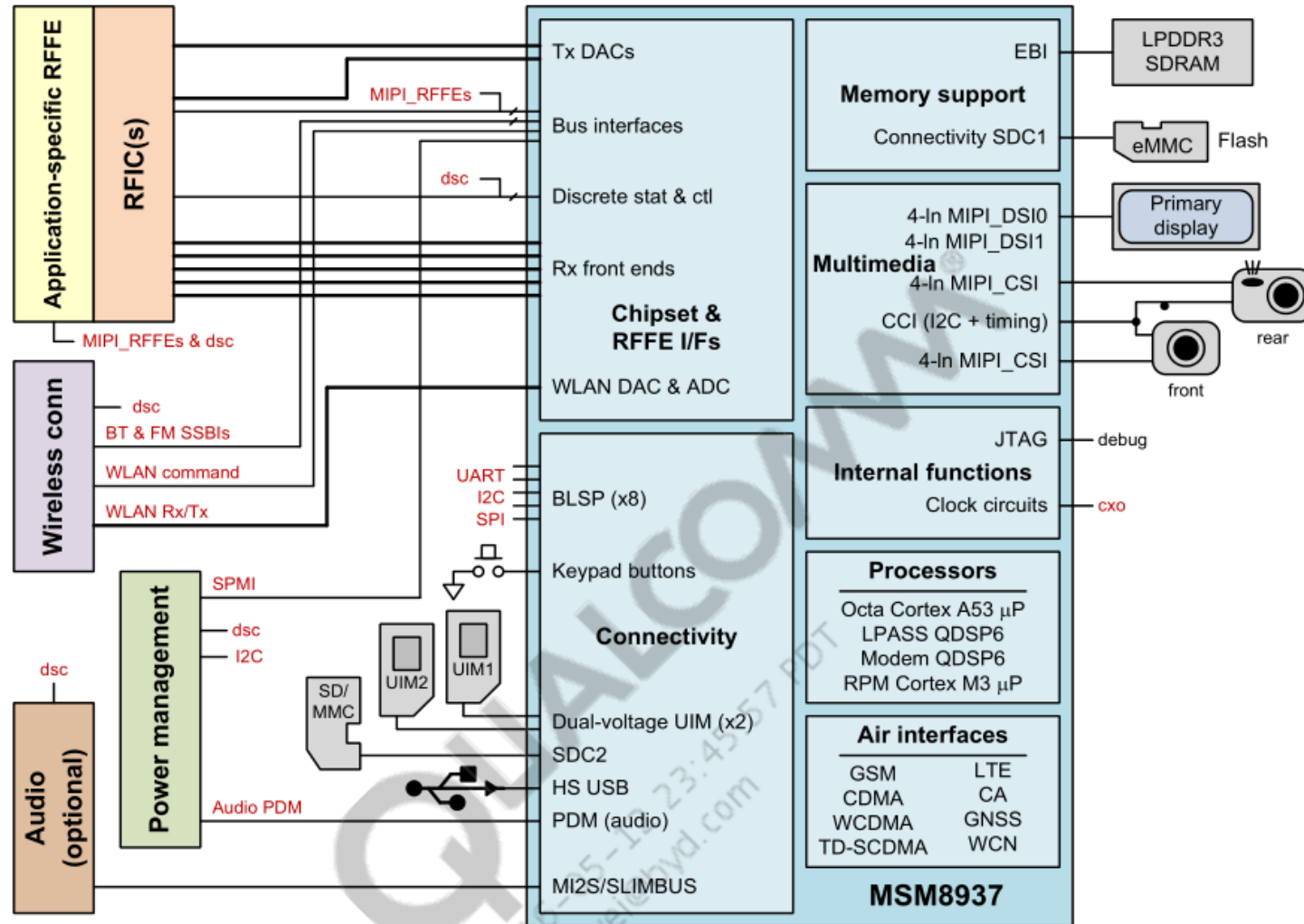


Figure 3-2. Power-on/off control sequence with XTAL by pressing PWRKEY



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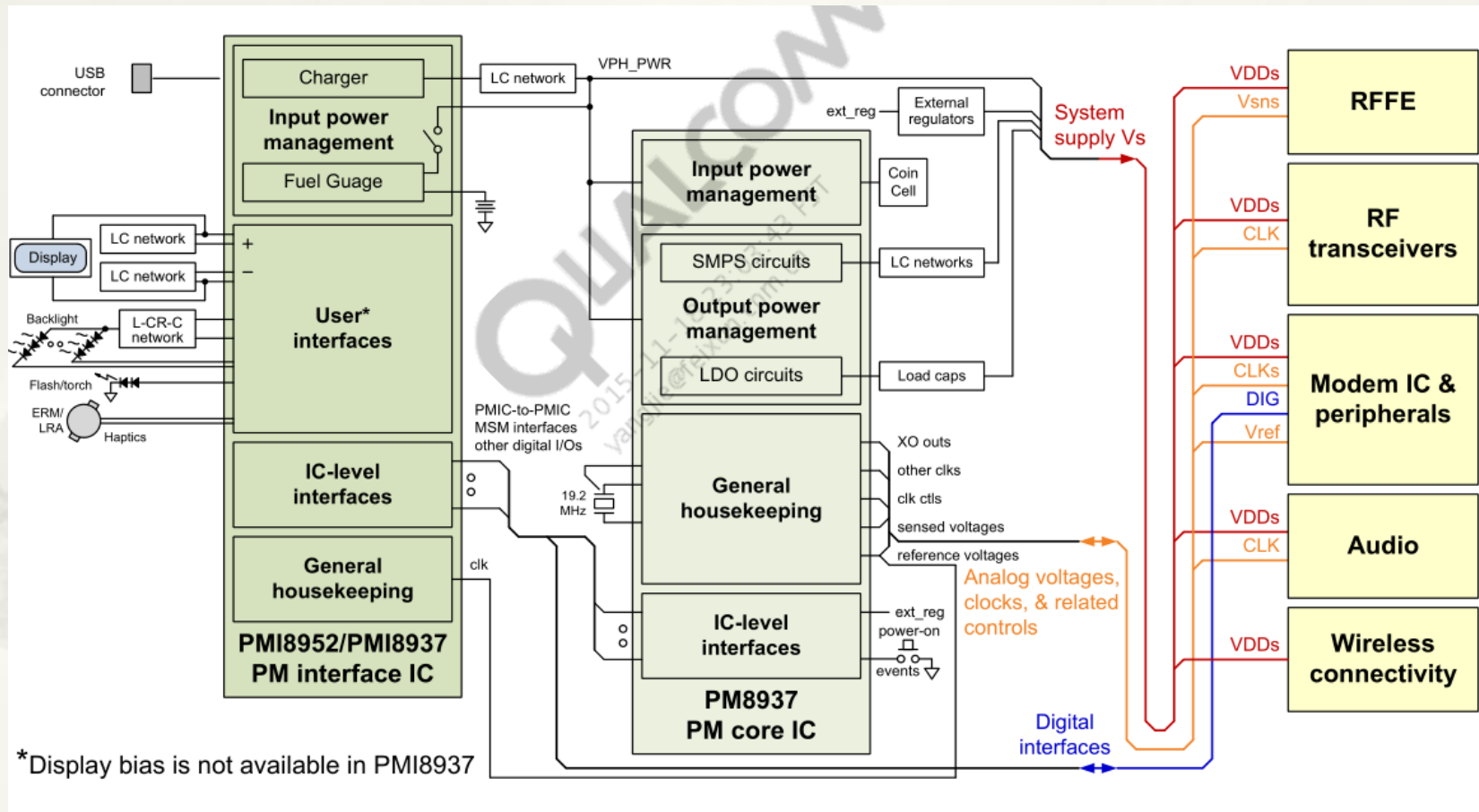
**Important devices and key test points**

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# I2C address

|             |      |            |                |            |                       |
|-------------|------|------------|----------------|------------|-----------------------|
| 加速度传感器      | \    | INVESENSE  | ICM-20608D     | I2C4       | Write:0xD0, Read:0xD1 |
| 地磁传感器       | \    | VTC        | AF8133J        | I2C4       | Write:0x3C, Read:0x3D |
|             |      | MEMSIC     | MMC3680KJ      | I2C4       |                       |
| 陀螺仪传感器      | \    | ST         | LSM6DS33       | I2C4       | Write:0xD4, Read:0xD5 |
|             |      | INVESENSE  | ICM-20608D     | I2C4       | Write:0xD0, Read:0xD1 |
| 三合一传感器      | \    | LITEON     | LTR-578ALS-039 | I2C4       | Write:0xA6 \Read:0xA7 |
| SAR 传感器     | \    |            |                |            |                       |
| 前摄像头        | 后摄8M | Omnivision | OV8856         | I2C1       | 0X20/0X21             |
|             |      | Hynix      | HI846          |            | 0X40/0X41             |
|             | 前摄8M | Hynix      | HI846          | I2C1       | 0X46/0X47             |
|             |      | Omnivision | OV8856         |            | 0X6C/0X6D             |
|             |      | Hynix      | HI1332         | I2C1       | 0X40/0X41             |
|             |      | Omnivision | OV13858        |            | 0X6C/0X6D             |
| 后摄Driver IC | 后摄8M | Giantec    | CN3937         | I2C0       | 0X18/0X19             |
|             |      | Giantec    | GT9762         |            | 0X18/0X19             |
|             | 后摄8M | Giantec    | GT24C64A       | I2C1       | 0XA0/0XA1             |
|             |      | Giantec    | GT24C64A       |            | 0XA0/0XA1             |
|             | 前摄5M | Giantec    | GT24C64AH      | I2C1       | 0XA8/0XA9             |
|             |      | Giantec    | GT24C64AH      |            | 0XA8/0XA9             |
| LCD         |      |            |                | GPIO10 ,11 |                       |
|             |      |            |                | GPIO10 ,11 |                       |
|             |      |            |                | GPIO10 ,11 |                       |
| 触摸屏         |      |            |                | GPIO10 ,11 |                       |

# Power management



# Buck Power management

| Function | Circuit type | Default V (V) <sup>1</sup> | Specified range (V) <sup>2</sup> (MSM8937) | Programmable range (V) | Rated current (mA) | Default on | Expected use (MSM8937)                            |
|----------|--------------|----------------------------|--------------------------------------------|------------------------|--------------------|------------|---------------------------------------------------|
| S1       | ULT-SMPS     | 1.225                      | 0.900–1.350                                | 0.375–1.5625           | 2000 <sup>3</sup>  | N          | MSM modem                                         |
| S2       | ULT-SMPS     | 1.225                      | 0.550–1.350                                | 0.375–1.5625           | 3000               | Y          | MSM core and graphics                             |
| S3       | HF-SMPS      | 1.288                      | 1.200–1.4125                               | 0.375–1.5625           | 2700               | Y          | Low-voltage LDOs (1, 2, 3, 19, and 23)            |
| S4       | ULT-SMPS     | 2.050                      | 1.800–2.050                                | 1.550–3.126            | 2500               | Y          | High-voltage LDOs (4, 5, 6, 7, 16, RFCLK, and XO) |
| S5       | FT-SMPS      | 1.225                      | 1.050–1.350                                | 0.350–1.355            | 3000               | Y          | MSM applications processor                        |
| S6       | FT-SMPS      | 1.225                      | 1.050–1.350                                | 0.350–1.355            | 3000               | N          | MSM applications processor                        |

# LDO Power management

|                  |               |                                                                                                                   |                                               |                        |                    |            |                                                                |
|------------------|---------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------------------------|--------------------|------------|----------------------------------------------------------------|
| L1               | NMOS LDO      | 1.000                                                                                                             | 1.000                                         | 0.375–1.5375           | 1200               | N          | RFICs                                                          |
| L2               | NMOS LDO      | 1.200                                                                                                             | 1.200                                         | 0.375–1.5375           | 1200               | Y          | LPDDR2/LPDDR3, MIPI CSI, and DSI                               |
| L3               | NMOS LDO      | 1.225                                                                                                             | 0.750–1.350                                   | 0.375–1.5375           | 1200               | Y          | VDDMX                                                          |
| L4               | PMOS LDO      | 1.800                                                                                                             | 1.800                                         | 1.750–3.3375           | 450                | N          | RFICs and GPS eLNA                                             |
| L5 <sup>4</sup>  | PMOS LDO      | 1.800                                                                                                             | 1.800                                         | 1.750–3.3375           | 500                | Y          | Most digital I/Os, MSM pad groups 3 and 7, LPDDR, and eMMC     |
| L6               | PMOS LDO      | 1.800                                                                                                             | 1.800                                         | 1.750–3.3375           | 300                | N          | MSM DSI PLL and OTP, camera, touchscreen, display, and sensors |
| L7               | PMOS LDO      | 1.800                                                                                                             | 1.800                                         | 1.750–3.3375           | 150                | Y          | MSM analog and PLLs, WCN XO, and PM baseband clock driver      |
| L8               | PMOS LDO      | 2.900                                                                                                             | 2.900                                         | 1.750–3.3375           | 600                | Y          | eMMC                                                           |
| L9               | PMOS LDO      | V <sub>out</sub> = 3.3 V for V <sub>BAT</sub> > 3.575 V;<br>V <sub>out</sub> = 3 V for V <sub>BAT</sub> < 3.575 V | 3.000–3.300                                   | 1.750–3.3375           | 600                | N          | WCN                                                            |
| L10              | PMOS LDO      | 2.8                                                                                                               | 2.800                                         | 1.750–3.3375           | 150                | N          | Sensors                                                        |
| L11 <sup>5</sup> | PMOS LDO      | 2.950                                                                                                             | 2.950                                         | 1.750–3.3375           | 800                | Y          | Micro SD                                                       |
| L12 <sup>4</sup> | PMOS LDO      | 2.950                                                                                                             | 1.800/2.950                                   | 1.750–3.3375           | 150                | Y          | MSM pad group 2 and SDC2                                       |
| L13              | PMOS LDO      | 3.075                                                                                                             | 3.075                                         | 1.750–3.3375           | 50                 | Y          | MSM USB and audio                                              |
| Function         | Circuit type  | Default V (V) <sup>1</sup>                                                                                        | Specified range (V) <sup>2</sup><br>(MSM8937) | Programmable range (V) | Rated current (mA) | Default on | Expected use (MSM8937)                                         |
| L14 <sup>5</sup> | PMOS LDO      | 1.800                                                                                                             | 1.800/2.925/3.050                             | 1.750–3.3375           | 50                 | N          | MSM pad group 5, dual-voltage UIM1, and NFC                    |
| L15 <sup>5</sup> | PMOS LDO      | 1.800                                                                                                             | 1.800/2.925/3.050                             | 1.750–3.3375           | 50                 | N          | MSM pad group 6 and dual-voltage UIM2                          |
| L16              | PMOS LDO      | 1.800                                                                                                             | 1.800                                         | 1.750–3.3375           | 5                  | N          | PMIC HKADC                                                     |
| L17              | PMOS LDO      | 2.850                                                                                                             | 2.850                                         | 1.750–3.3375           | 600                | N          | Camera, display, and touchscreen                               |
| L18              | PMOS LDO      | 2.700                                                                                                             | 2.700                                         | 1.750–3.3375           | 150                | N          | QTI RF front-end                                               |
| L19              | NMOS LDO      | 1.350                                                                                                             | 1.350                                         | 0.375–1.5375           | 1200               | N          | MSM analog, WCN, and WGR                                       |
| L20              | Low-noise LDO | 1.74                                                                                                              | 1.74                                          | 1.74–3.3375            | 5                  | Y          | PMIC XO circuits                                               |
| L21              | Low-noise LDO | 1.74                                                                                                              | 1.74                                          | 1.74–3.3375            | 5                  | Y          | PMIC RF clock buffers                                          |
| L22              | PMOS LDO      | 2.800                                                                                                             | 2.800                                         | 1.750–3.3375           | 300                | N          | Camera – analog                                                |
| L23              | NMOS LDO      | 1.3                                                                                                               | 1.2                                           | 0.375–1.5375           | 300                | N          | Camera – digital                                               |



# Charge switch machine timing

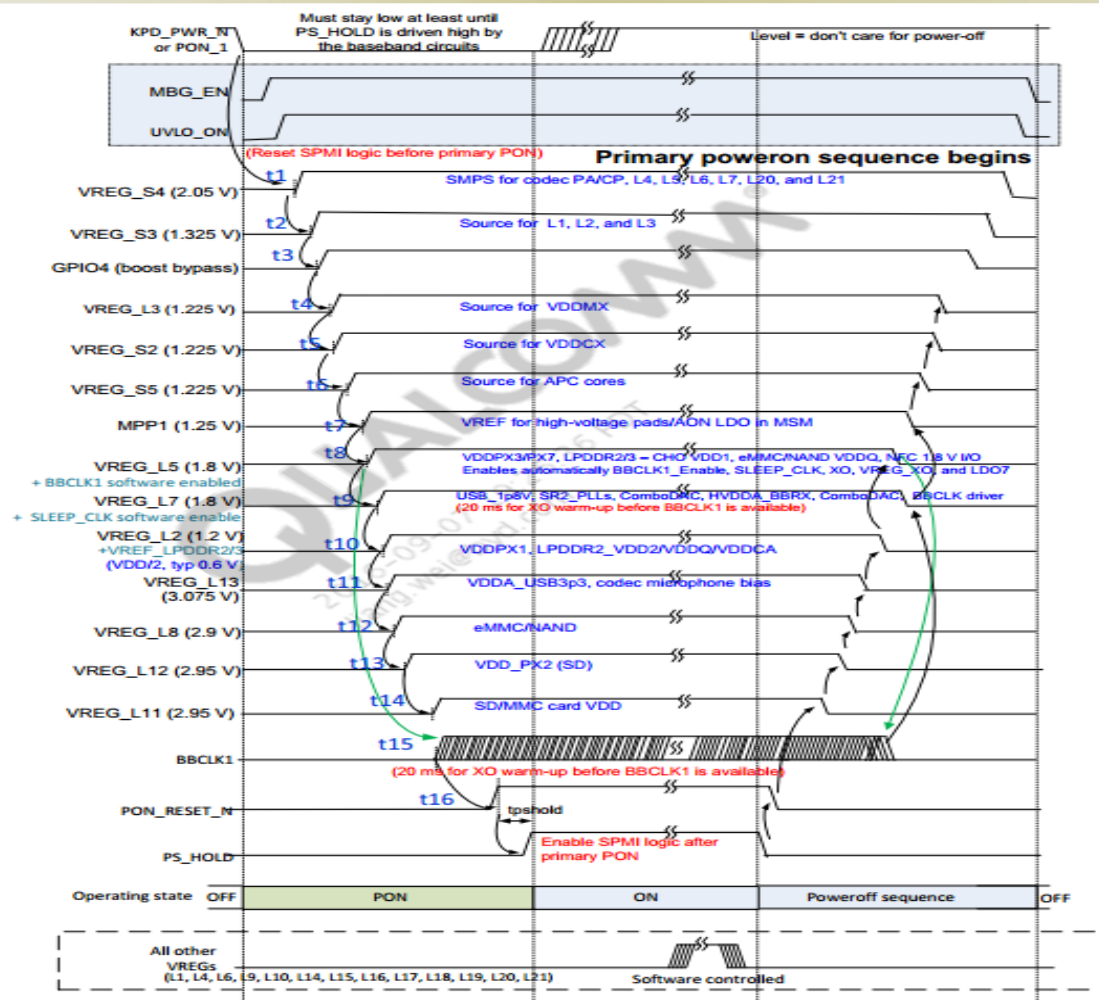


Figure 3-8 PM8937/PM8940 power-on sequence

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**Top view of the board**

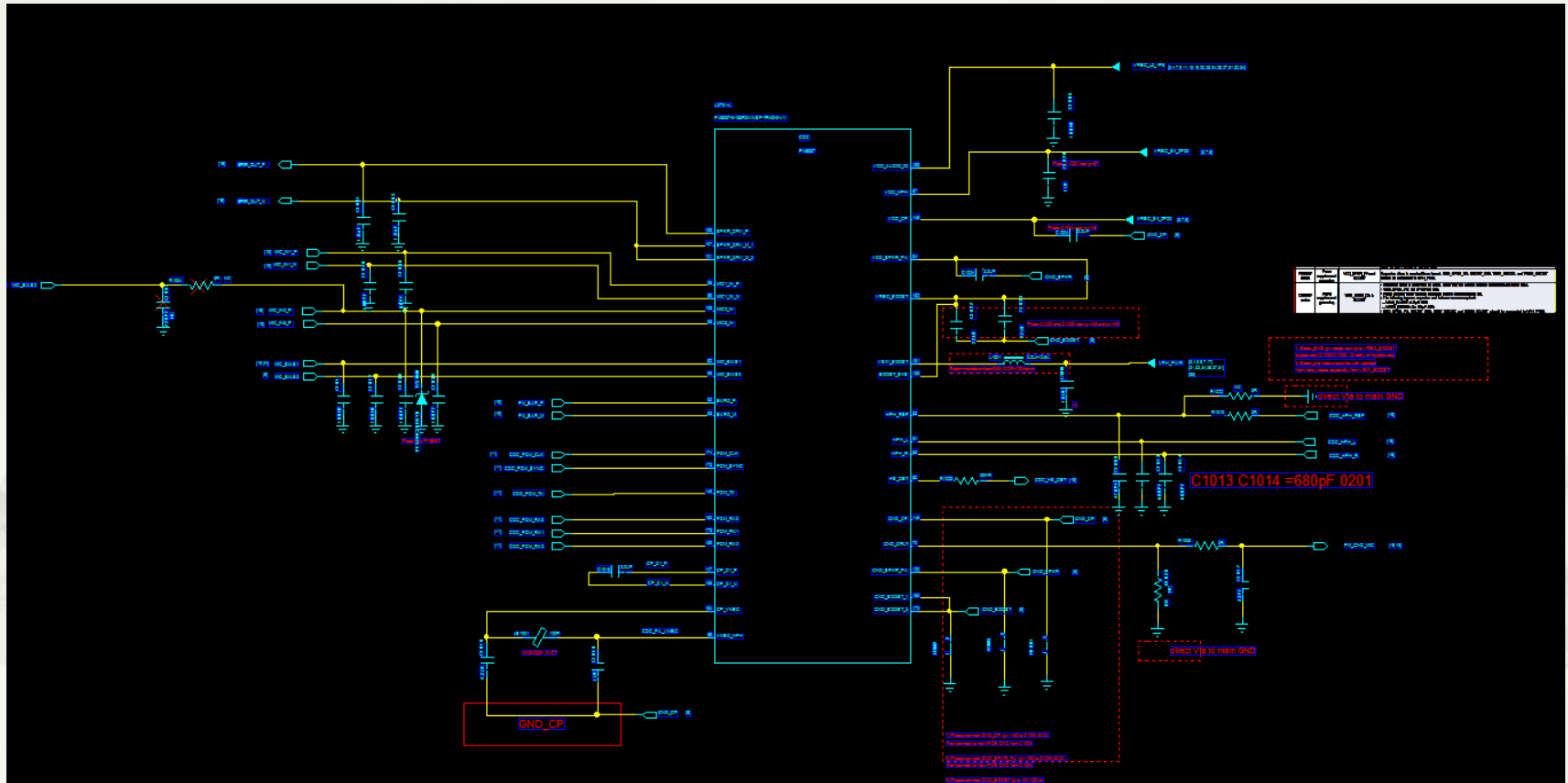
**System Block Diagram**

**I2C address and power management**

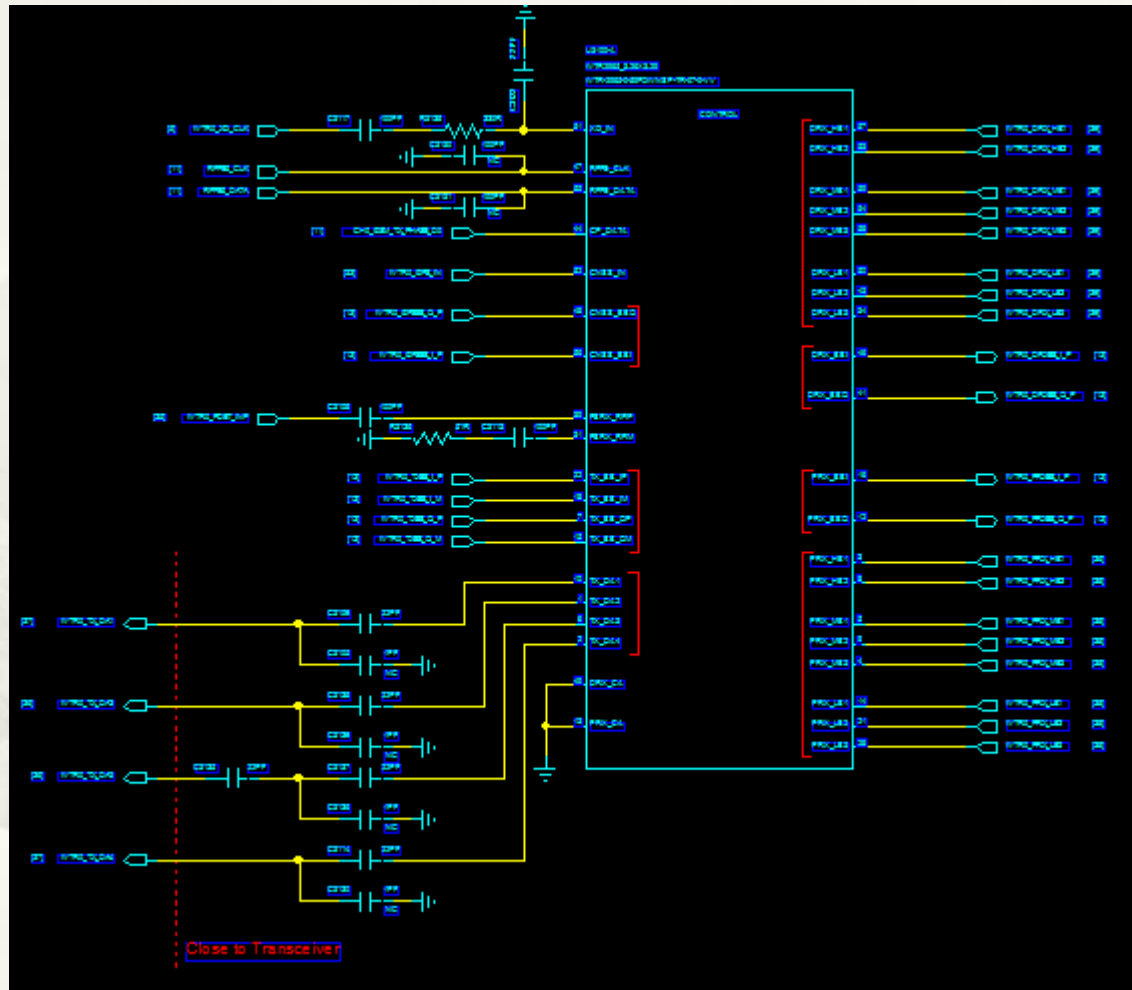
**Important devices and key test points**

**Maintenance methods**

**Power supply : PM8937 U0701**

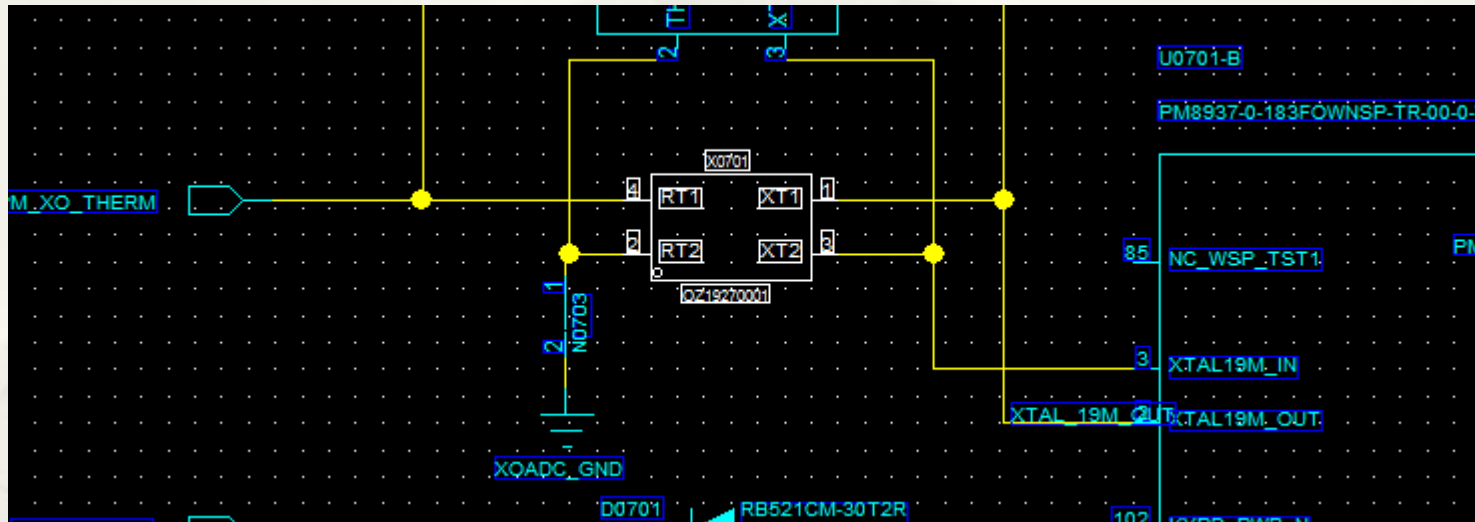


# Transceiver WTR2965 U3101



# 19MHz Crystal section X0701

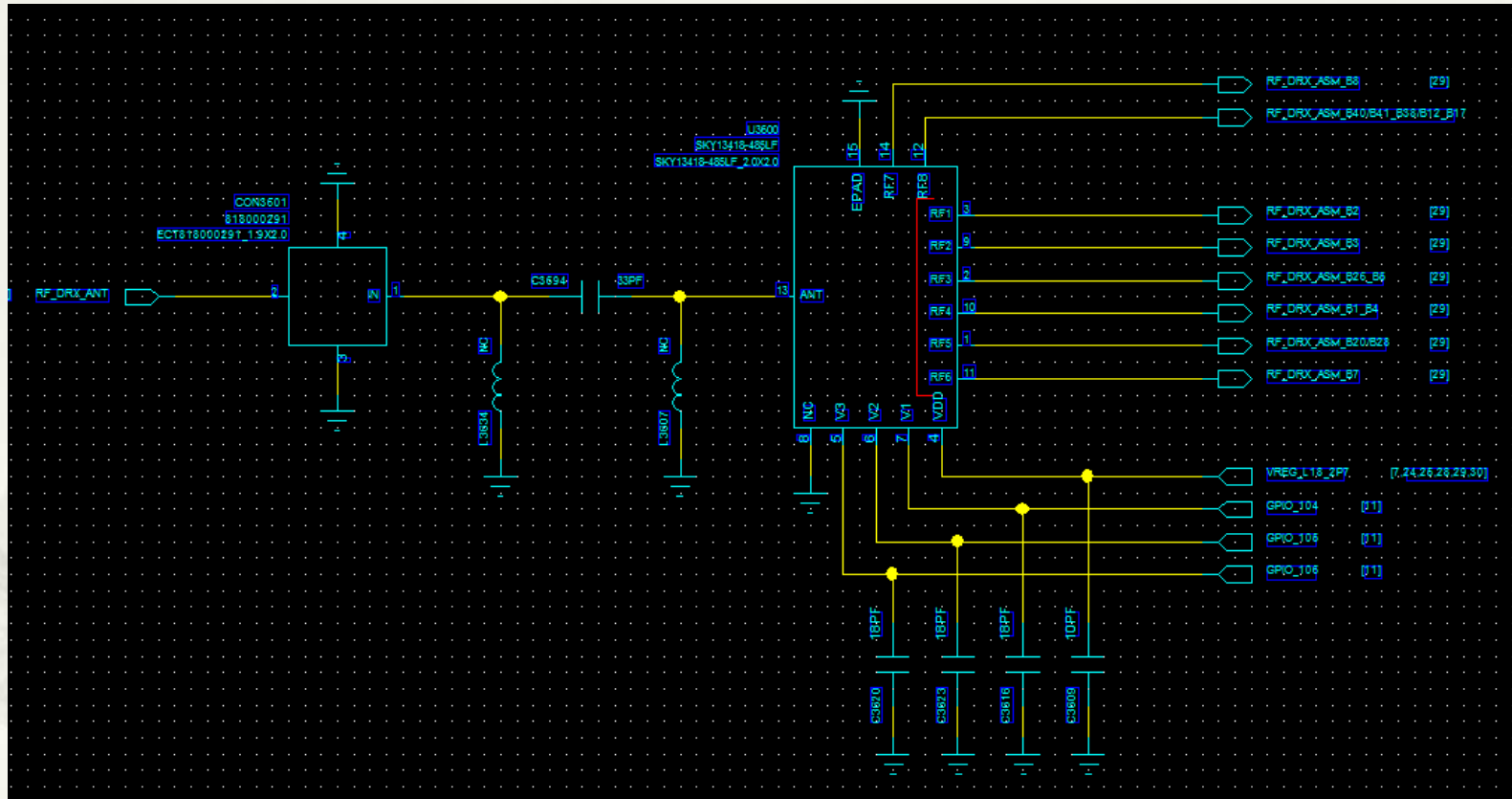
The output wave of Crystal looks like sine wave with amplitude 1V, frequency 19MHz



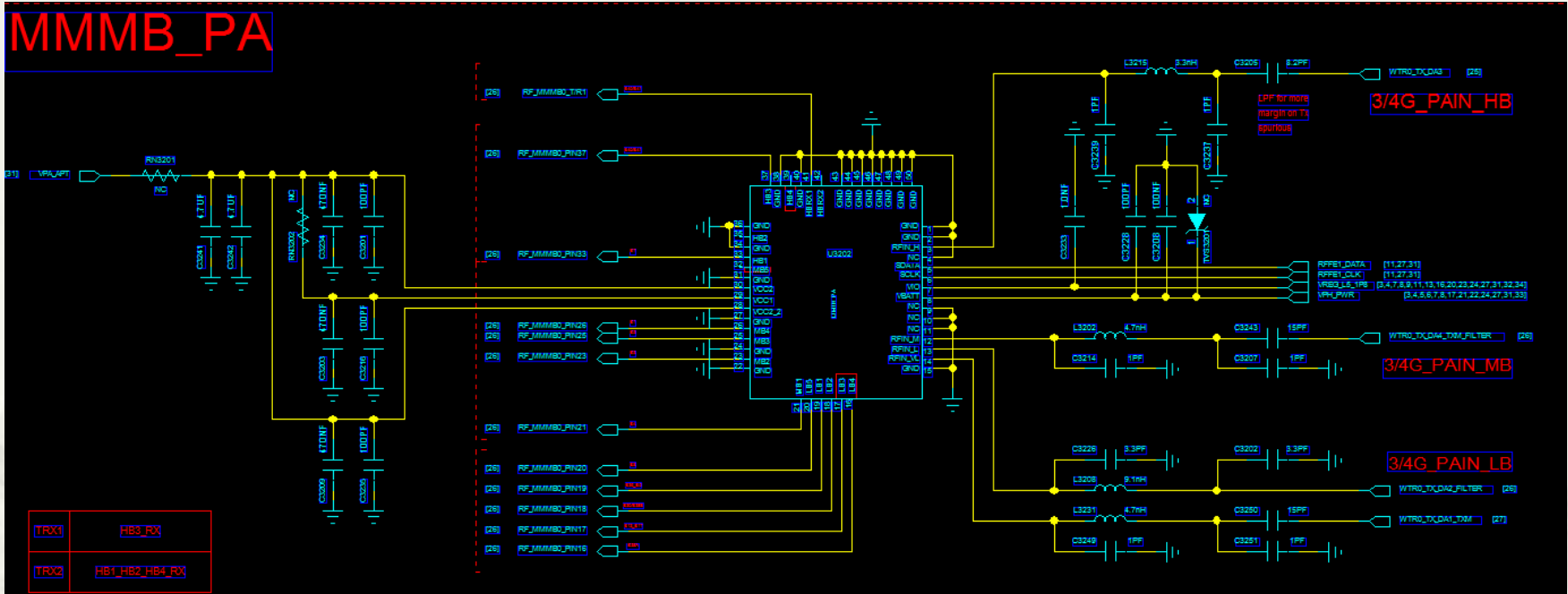
The schematic diagram illustrates the SWTP (Switching Transistor Power) section of a 2G GSM mobile phone. The circuit is powered by a 3.5V battery and a 3.5V regulator. The power supply section includes a 3.5V battery, a 3.5V regulator, and various passive components like capacitors and inductors. The 2G GSM module is connected to the power supply and the SWTP section. The SWTP section includes a 2G\_PAIN\_HB (2G Pain High Band) and a 2G\_PAIN\_LB (2G Pain Low Band) section. The SWTP section is color-coded with red and blue lines, and includes labels for components like SWTP, 2G\_PAIN\_HB, and 2G\_PAIN\_LB.



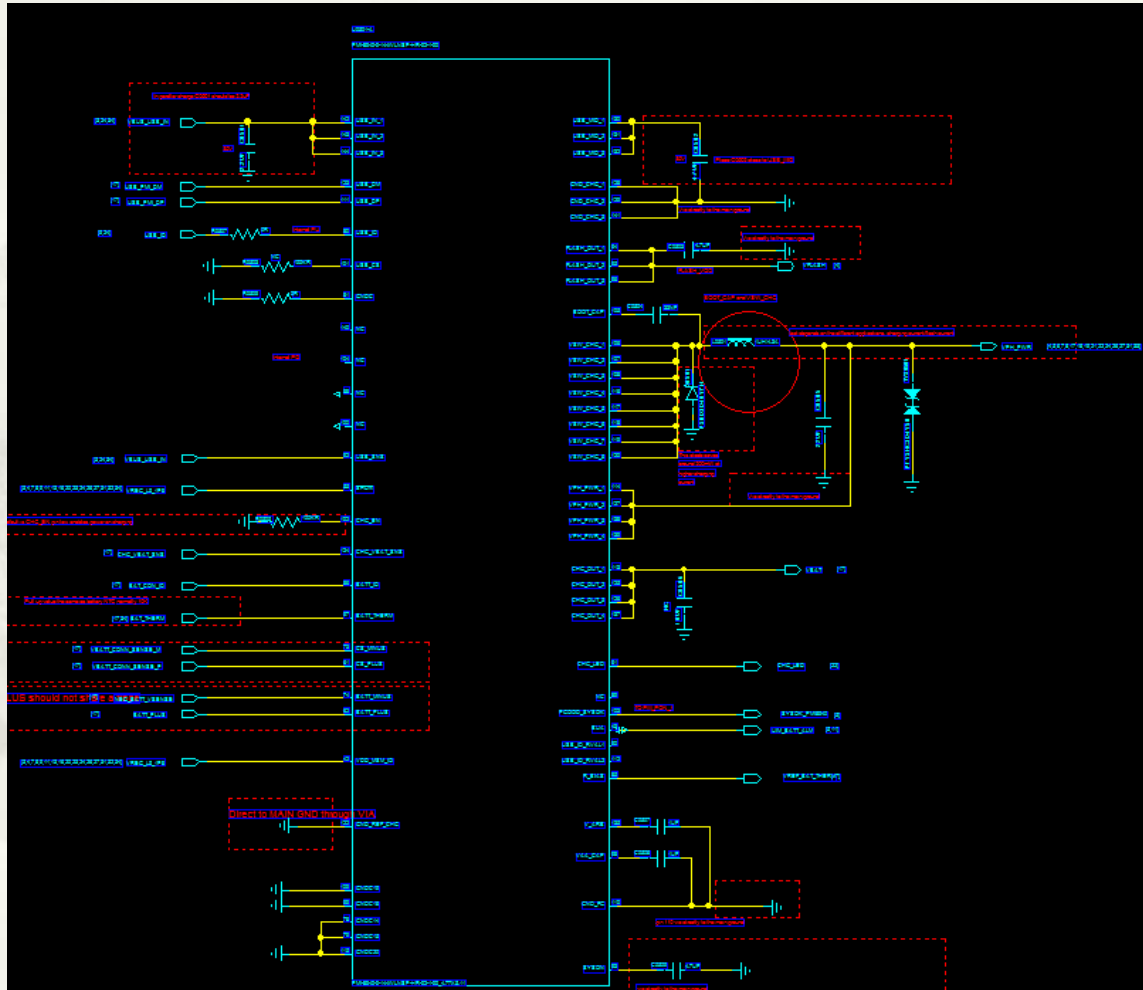
# SP8T U3600



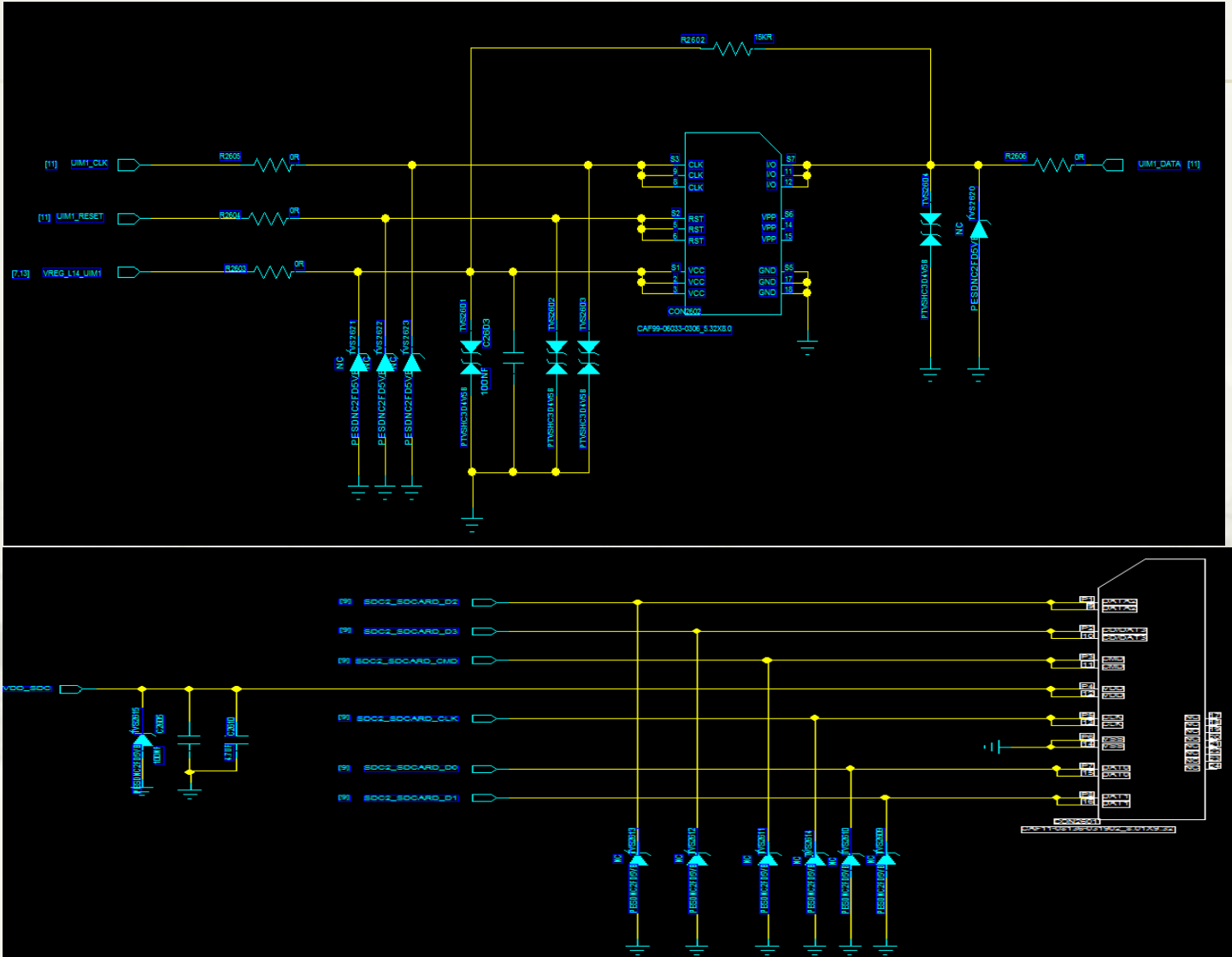
# MMPA U3202



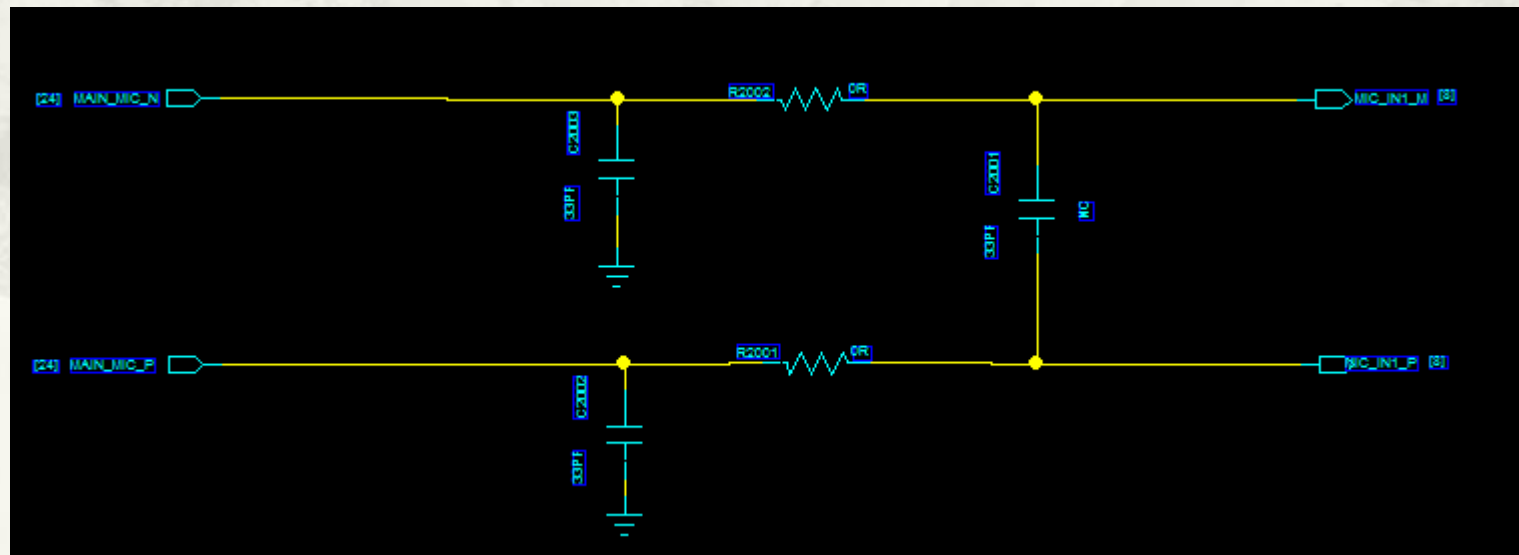
# Charge management PMI8940 U0501



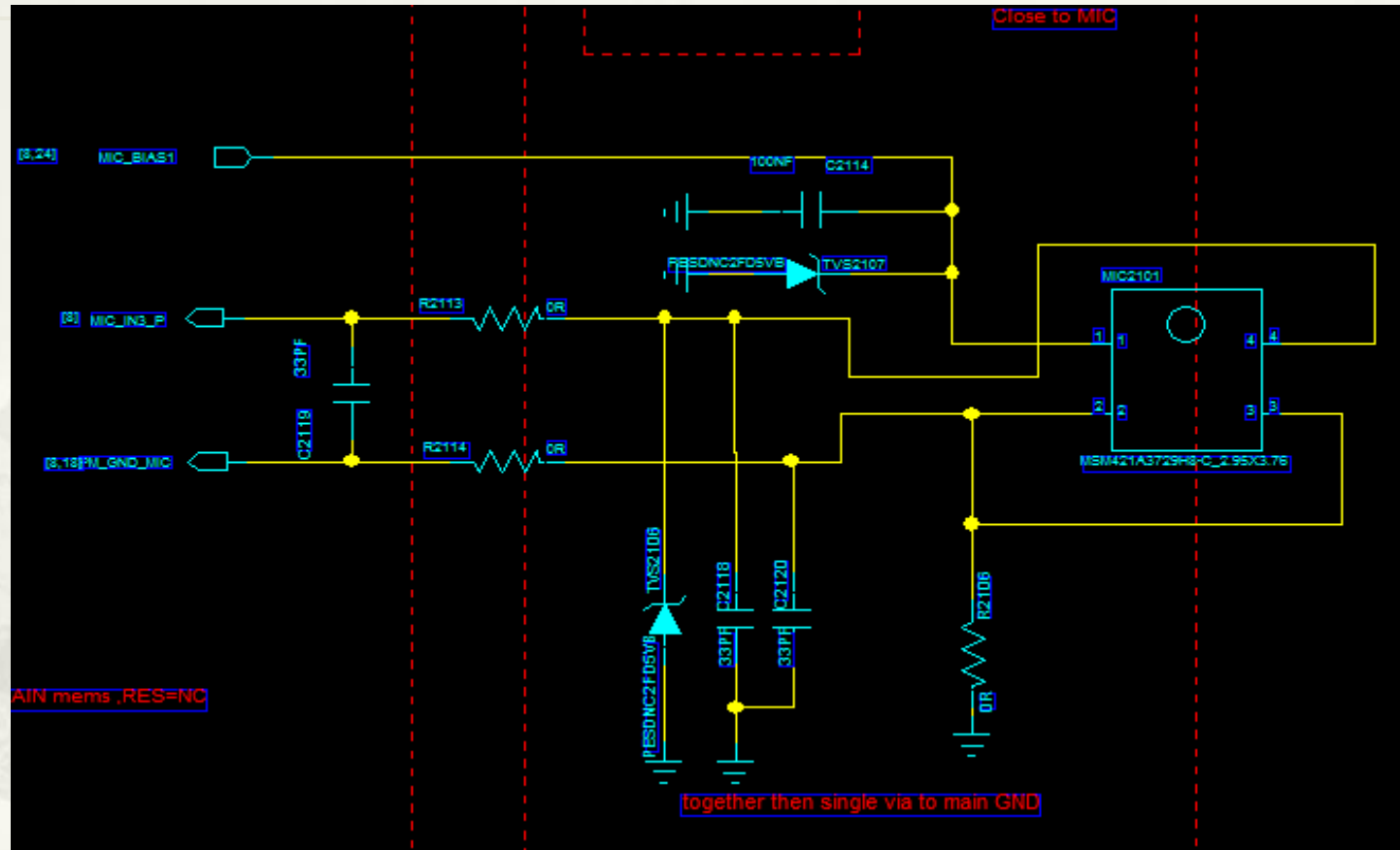
# SIM & TF socket con2602 & con2603 con2601



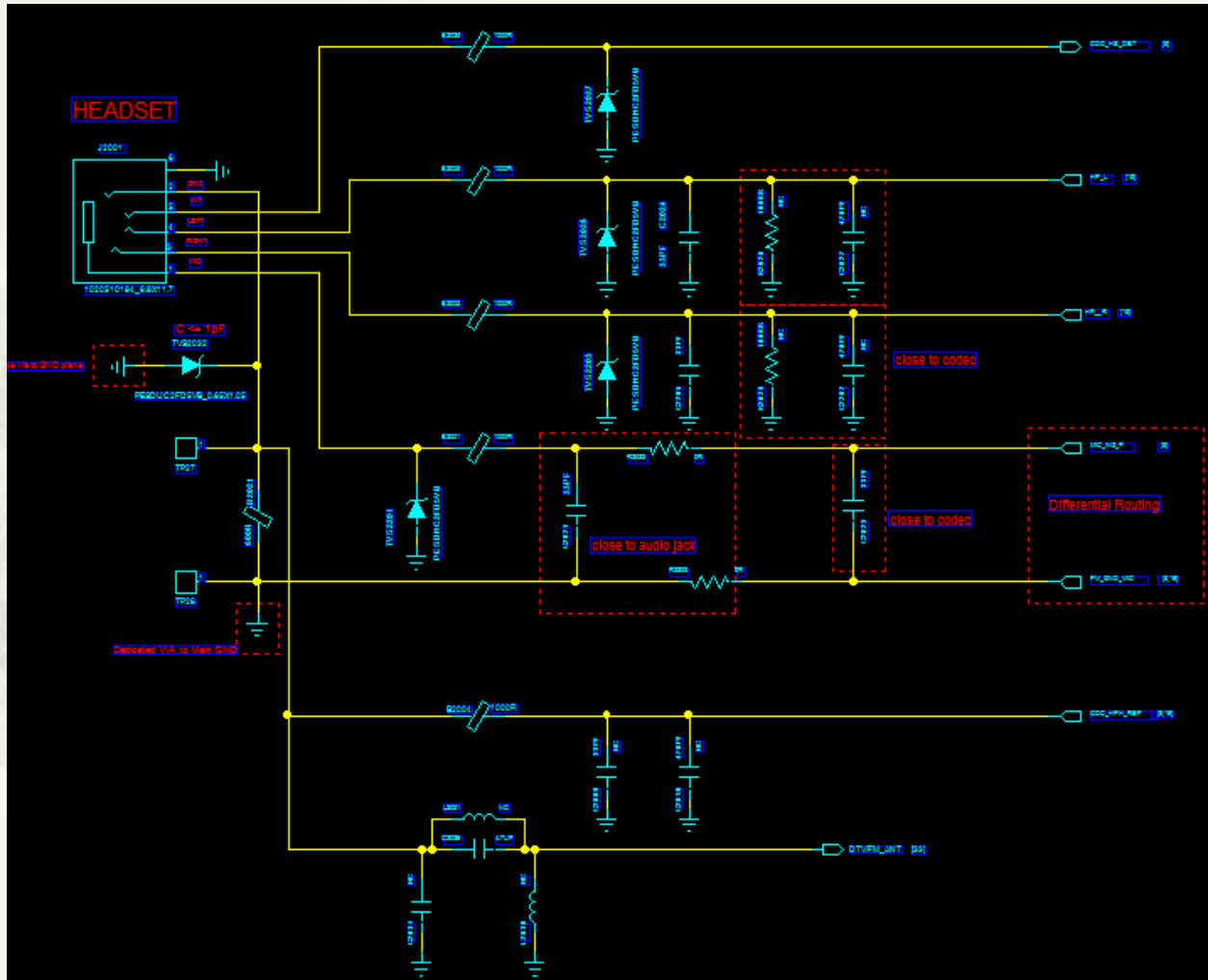
# Main MIC



# Second MIC

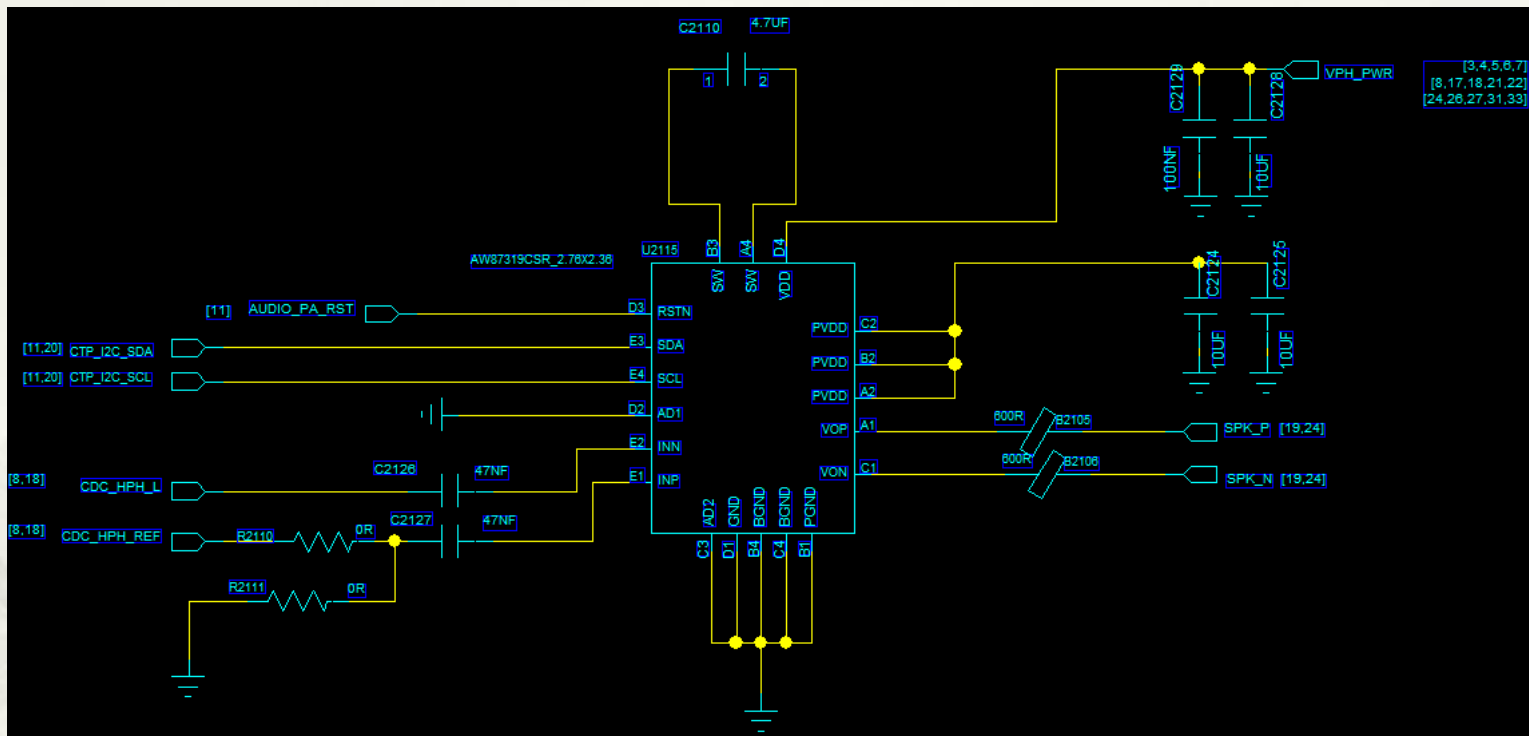


## Headset Jack section J2001

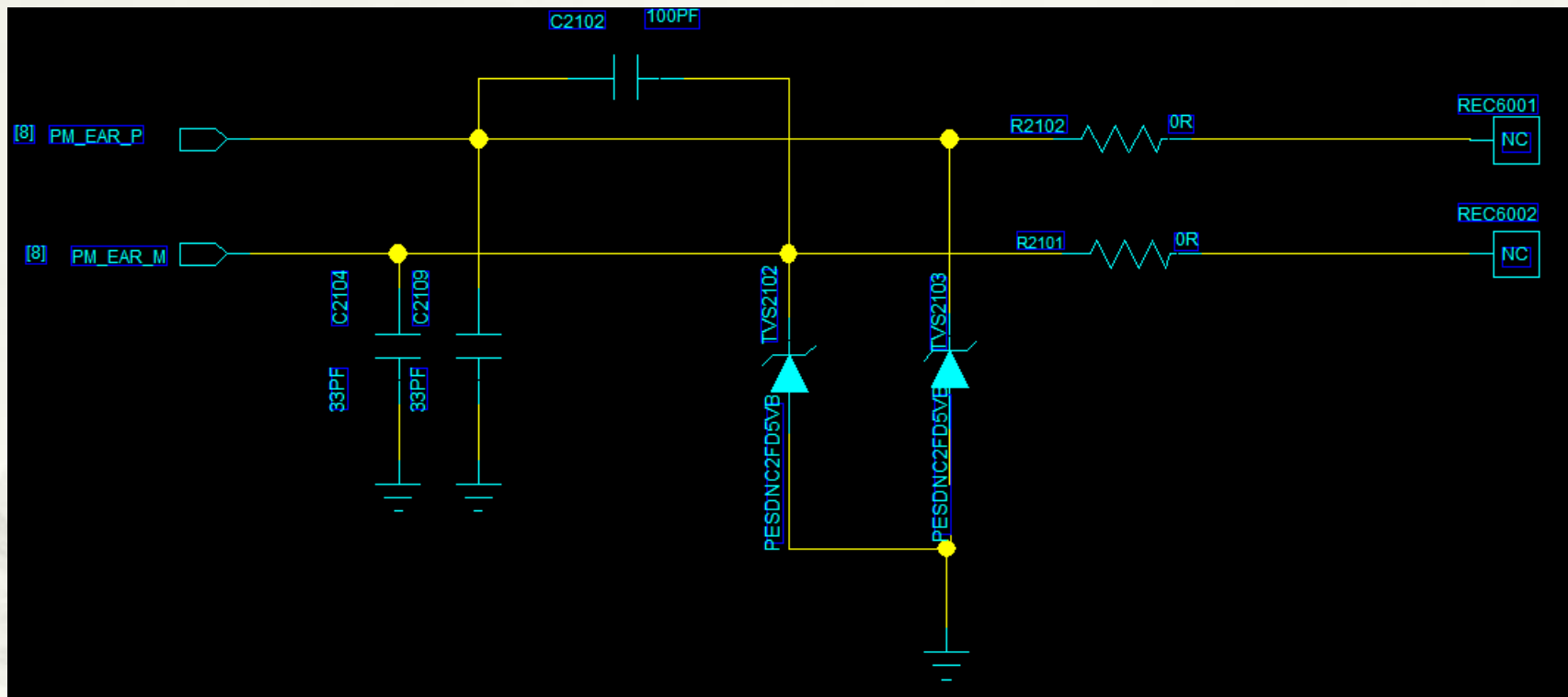




# Speaker PA AW87329 U2115



# Receiver section



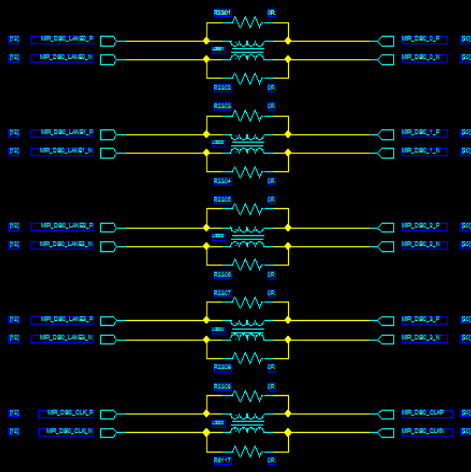
The image displays a series of circuit diagrams for various sensors and modules, organized into a grid. The diagrams include:

- Pressure Sensors:**
  - Pressure Sensor - 1 (Top Left)
  - Pressure Sensor - 2 (Top Right)
  - Pressure Sensor - 3 (Middle Left)
  - Pressure Sensor - 4 (Middle Right)
  - Pressure Sensor - 5 (Bottom Left)
  - Pressure Sensor - 6 (Bottom Right)
- Temperature Sensors:**
  - Temperature Sensor - 1 (Top Left)
  - Temperature Sensor - 2 (Top Right)
  - Temperature Sensor - 3 (Middle Left)
  - Temperature Sensor - 4 (Middle Right)
  - Temperature Sensor - 5 (Bottom Left)
  - Temperature Sensor - 6 (Bottom Right)
- Camera Modules:**
  - Camera - 1 (Top Left)
  - Camera - 2 (Top Right)
  - Camera - 3 (Middle Left)
  - Camera - 4 (Middle Right)
  - Camera - 5 (Bottom Left)
  - Camera - 6 (Bottom Right)
- Table of Contents:**
  - 1. INTRODUCTION
  - 2. THEORY
  - 3. COMPONENTS
  - 4. CIRCUIT DIAGRAM
  - 5. RESULTS
  - 6. CONCLUSION
  - 7. REFERENCES

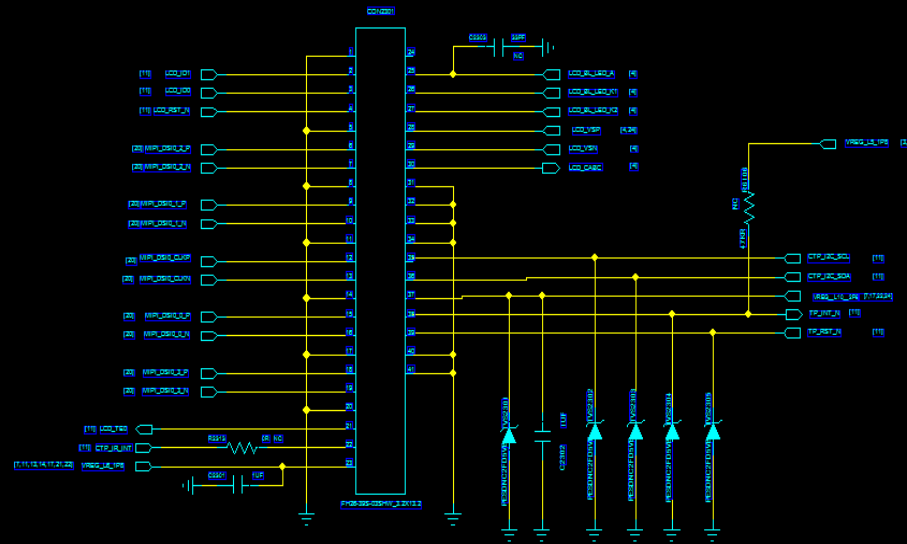
# LCM section con2301

## Main LCM

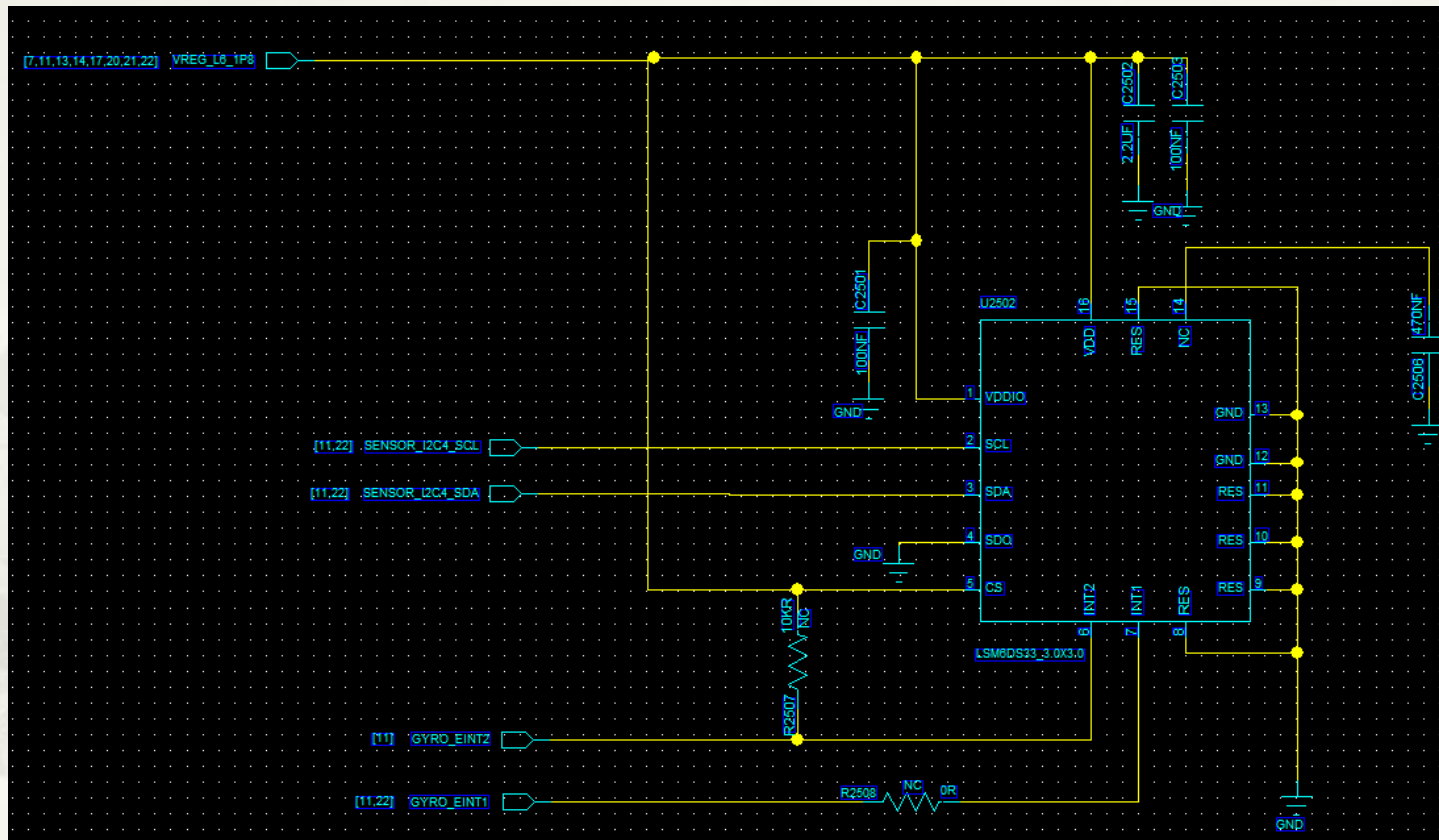
OR与共模EMI Share same pad



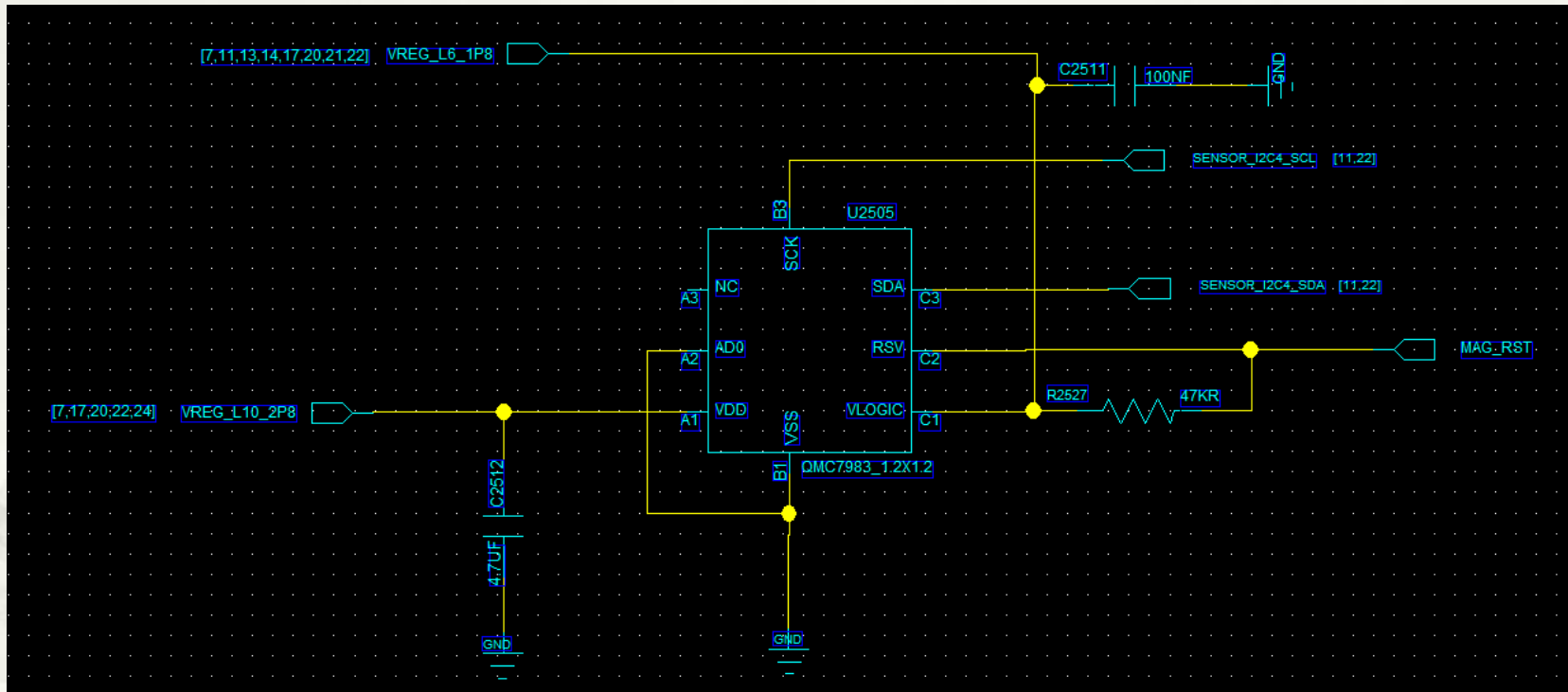
Yassy: ID0=0, ID1=0  
TXD: ID0=1, ID1=1



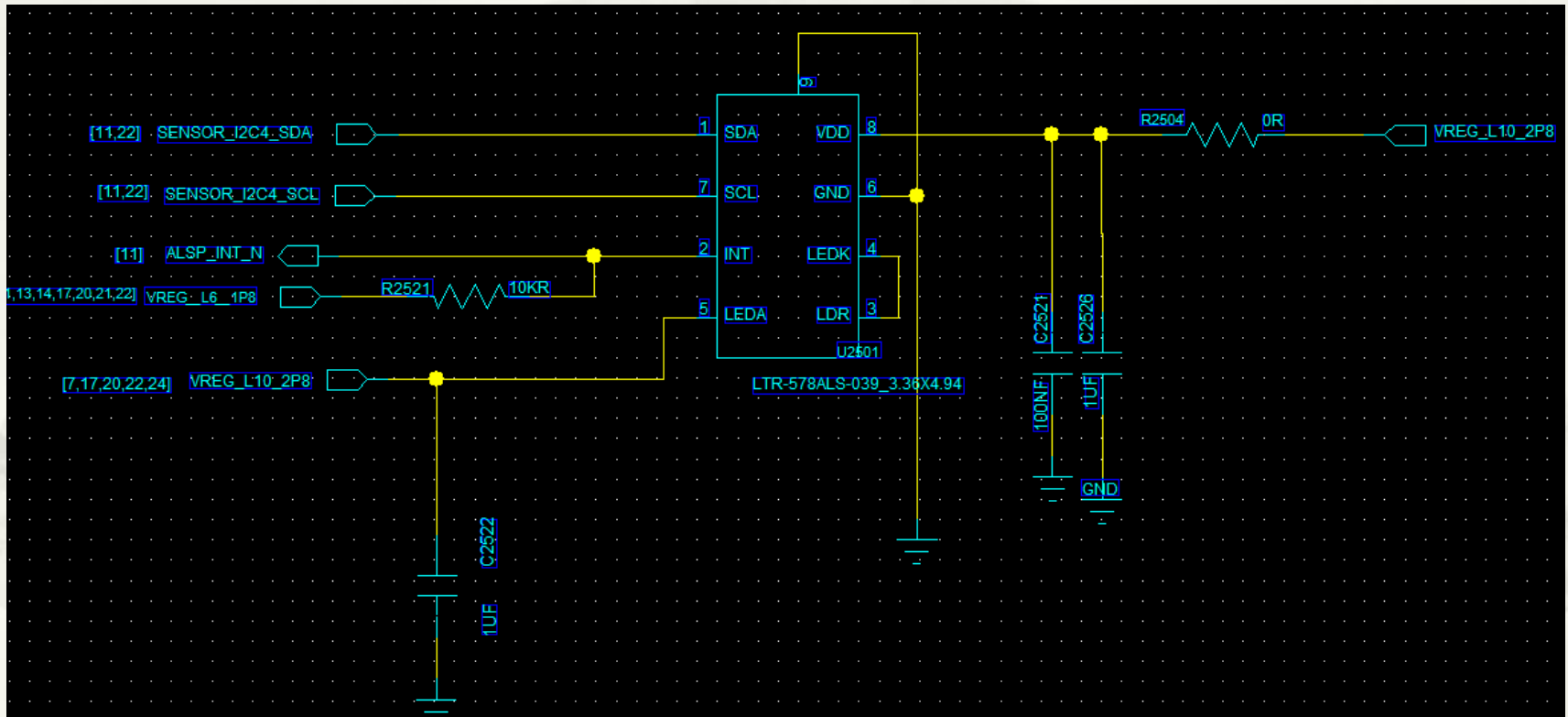
# G-sensor section U2502



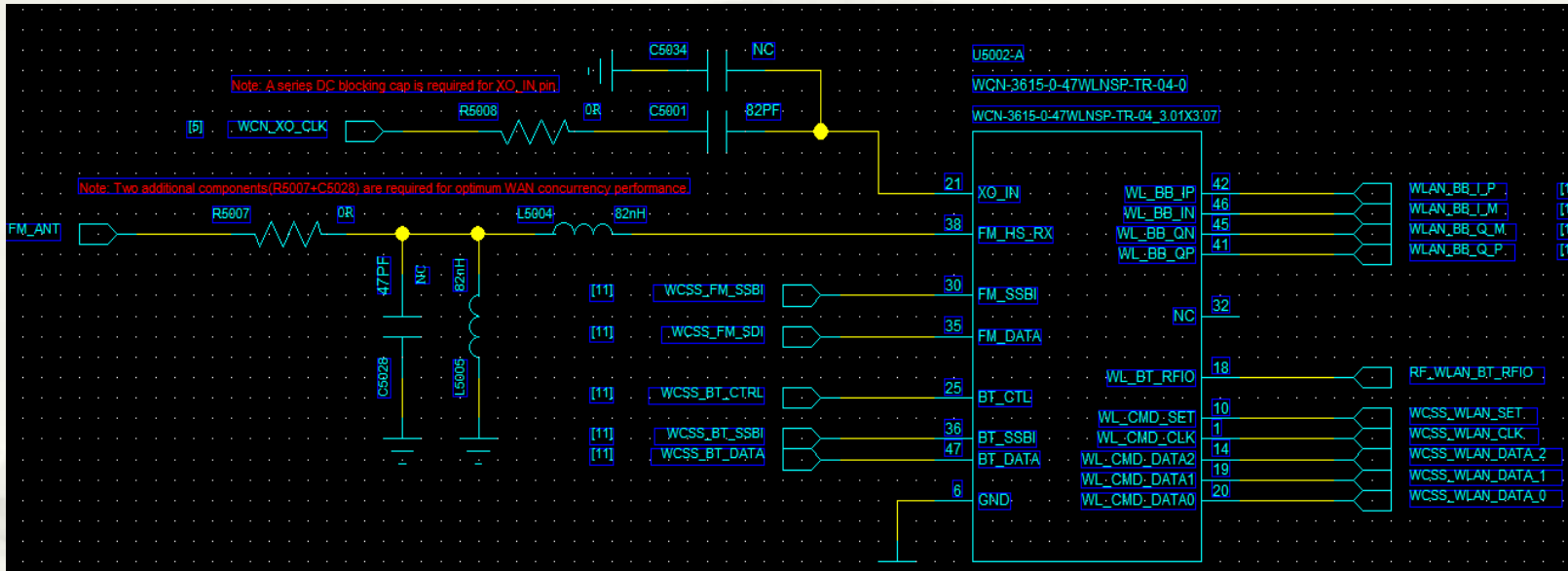
# M-sensor section U2505



# Ambient Light Sensor & Proximity Sensor section U2501

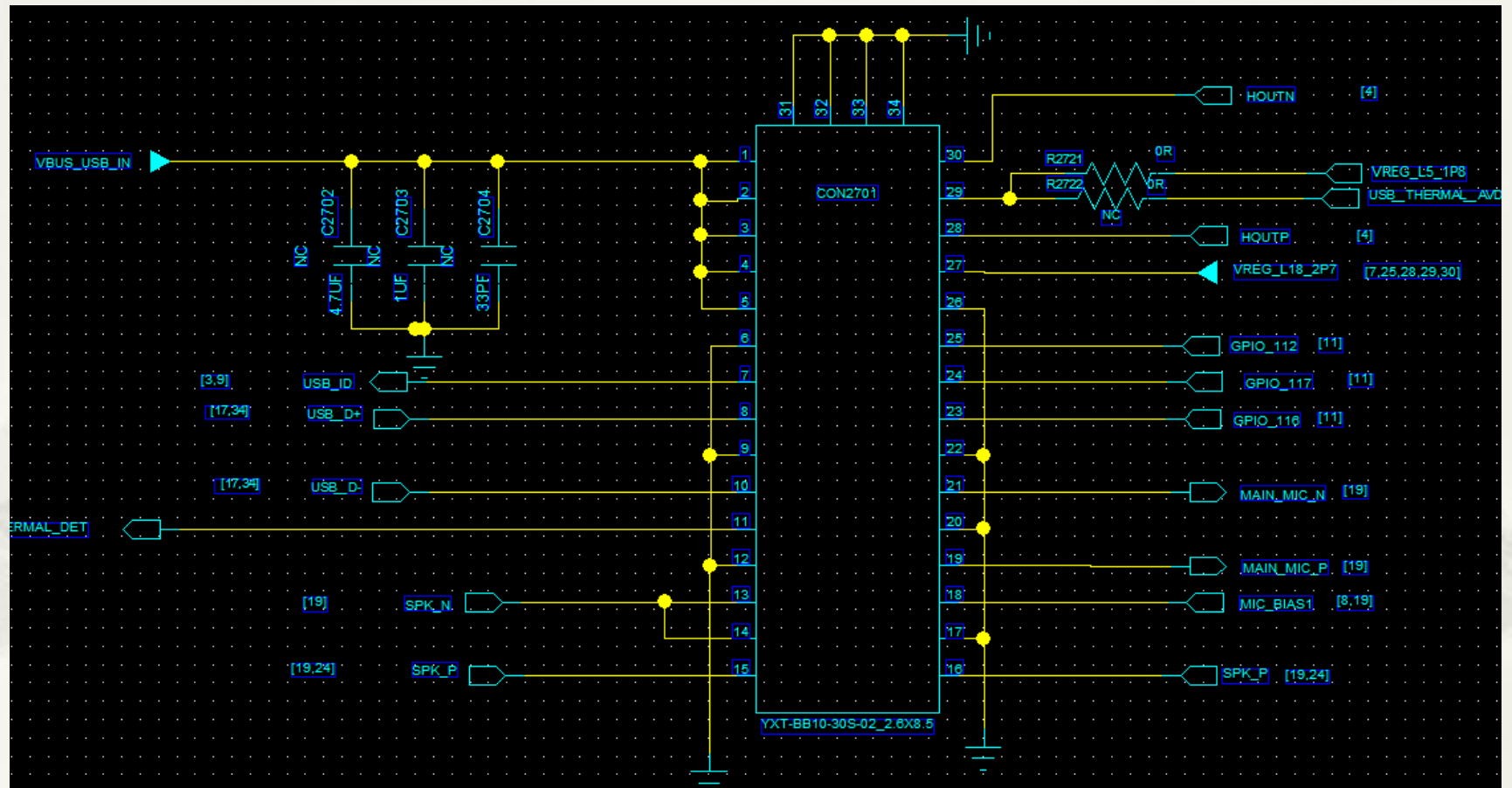


# GPS/WIFI/FM/Bluetooth WCN-3615

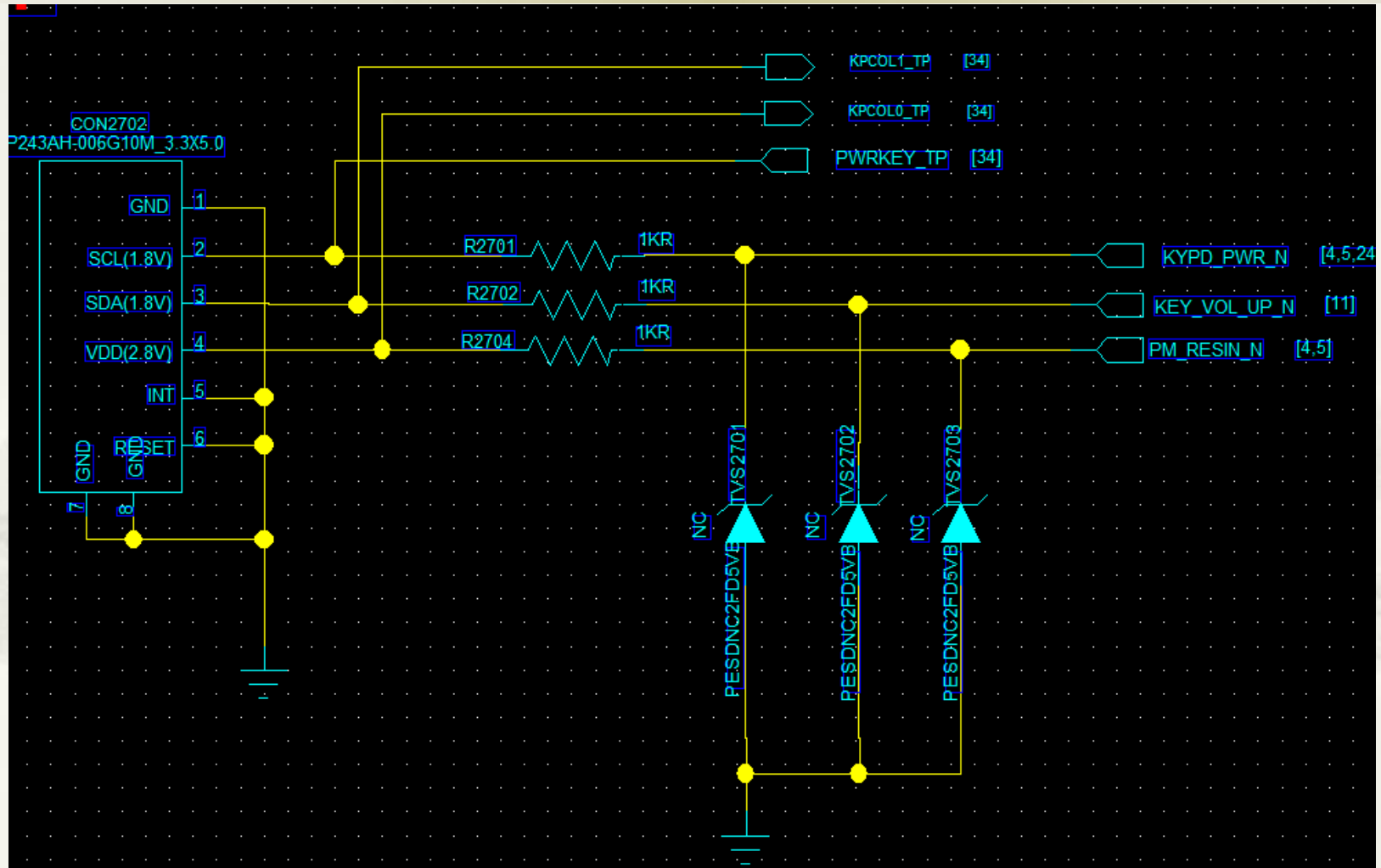




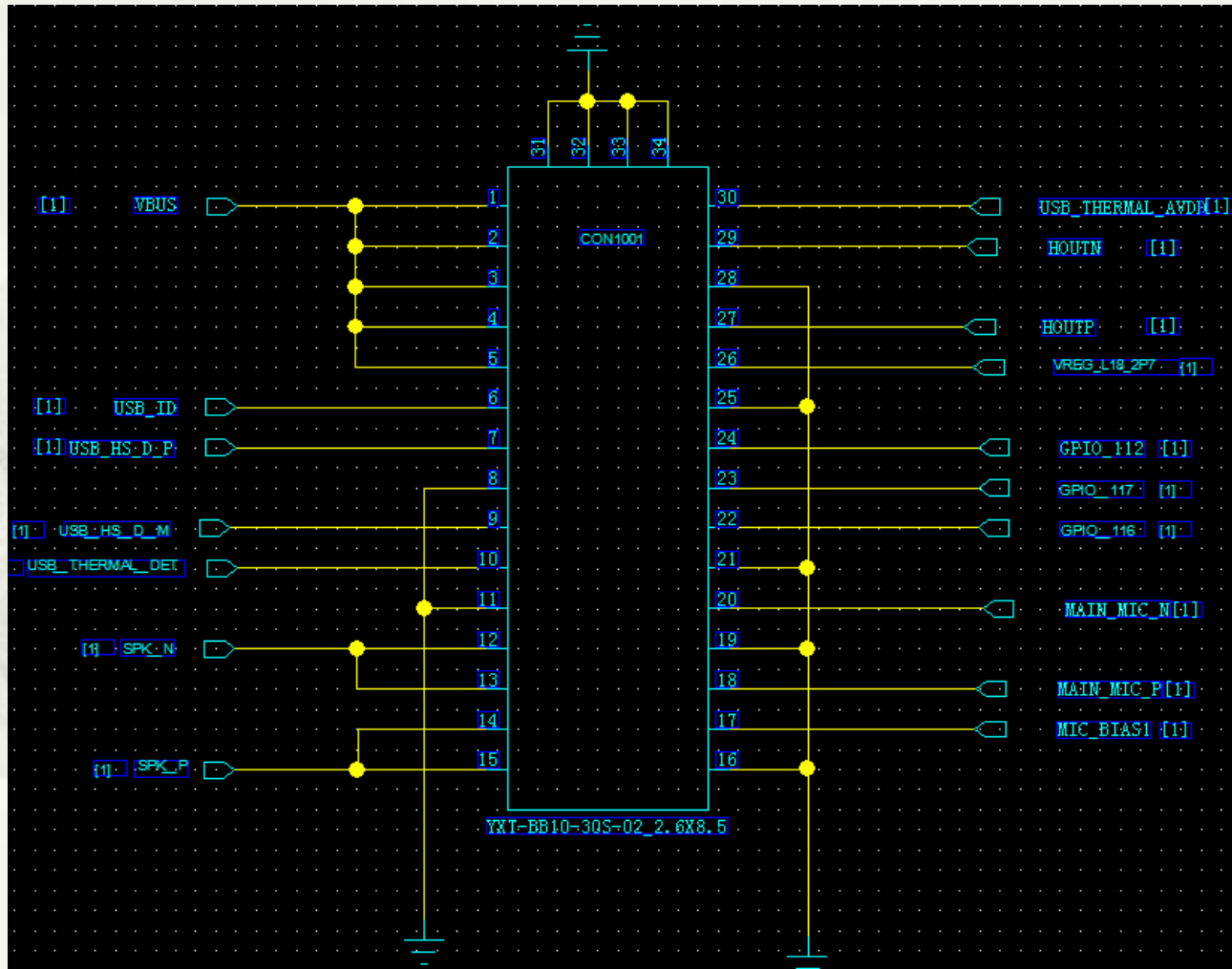
# To sub board CON2701



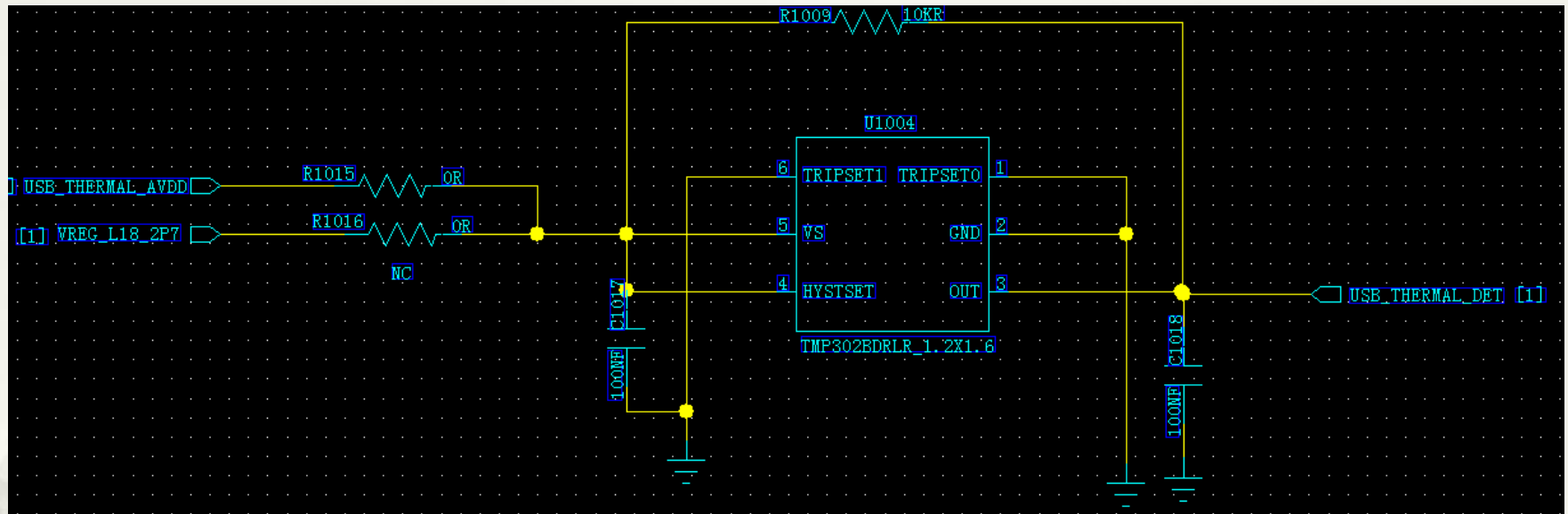
# Side key section CON2702



# Sub board



# Thermal Alert



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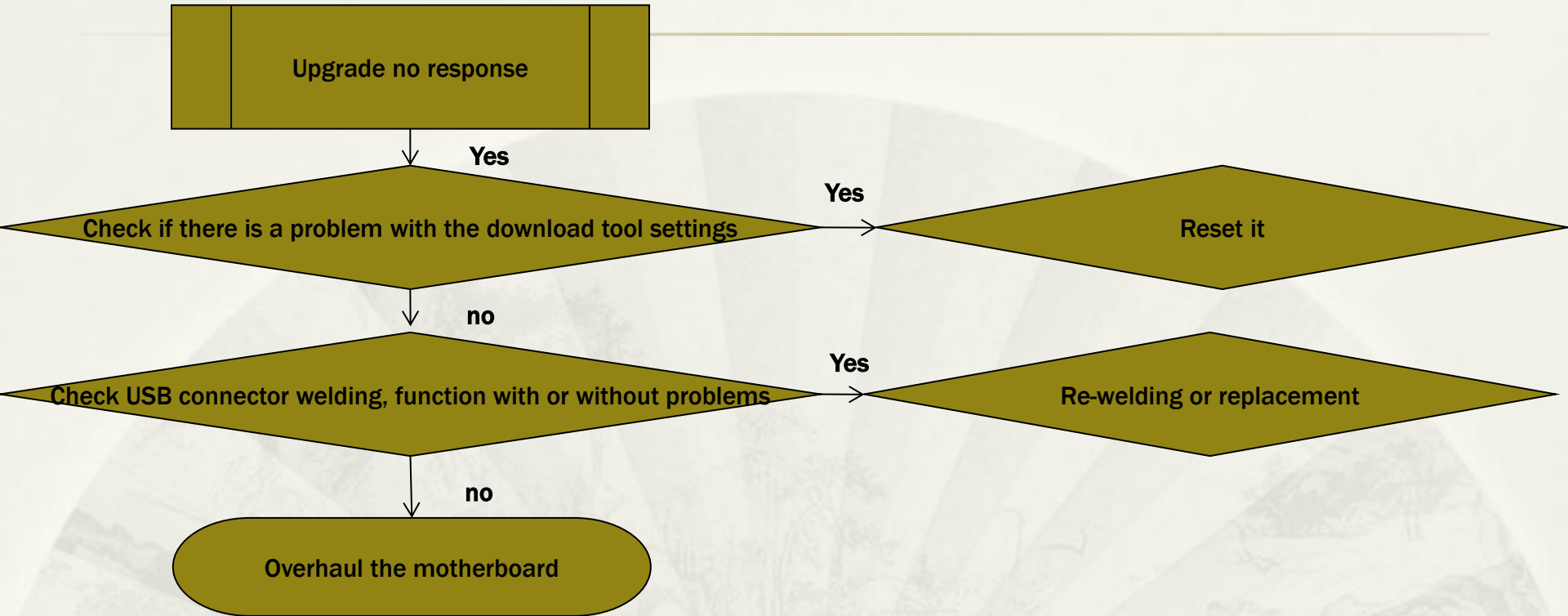
**System Block Diagram**

**I2C address and power management**

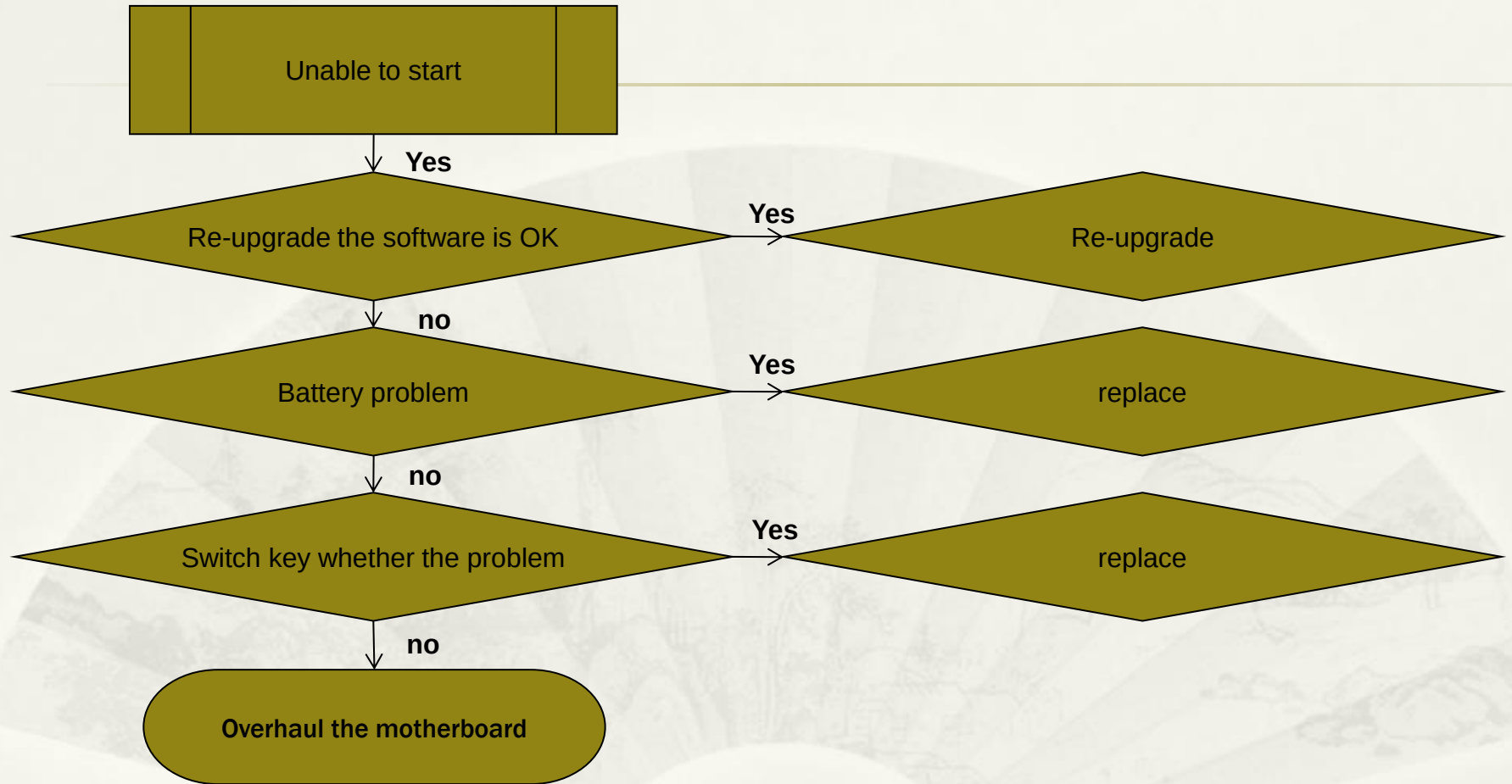
**Important devices and key test points**

**Maintenance methods**

# 1. Failure to fail analysis



## 2. Can not boot failure analysis



## 2.1 Can not boot failure analysis \_ power voltage and current abnormalities

- \* With a multimeter or oscilloscope to measure the battery cell P + voltage is low, and the power is too low, the voltage is lower than 3.4V:
- \* - charge, to be greater than 3.6V voltage, re-start the boot action .
- \* Battery P + voltage is greater than 3.6V, but the power leakage current :
  - Make sure TVS1901 is abnormal ;
  - Check whether the charging circuit U0501 is abnormal;
  - To confirm whether the radio frequency PA U3301, U3202 is abnormal;
  - To confirm U2501、 U2502、 U2505 is abnormal;
  - CheckPMIC U0701 is abnormal;
  - Check CON2101 pin1 or short circuit ;
  - Above devices in addition to PMIC U0701 other than , Other bit number removal does not affect boot , Analysis can be used to exclude the removal method and then boot test



## 2.2 Can not boot failure analysis \_ boot process current abnormal

The VBAT voltage is high with a multimeter or oscilloscope, but the current is not normal during startup

- To confirm whether there is a chip fever
- Check that the PMIC voltage output is normal
- Make sure the CPU is abnormal

## 2.3 Can not boot failure analysis clock

- \* Measure the VBAT voltage with a multimeter or oscilloscope is high
- \* --Make sure that the output voltage of PMIC U0701 is abnormal, There is an exception is welding or replacement PMIC further confirmed, if no exception is the next step
- \* --ConfirmX0701 output clock is normal
- Confirm19.2Mhz clock U3102 output 26M clock is normal

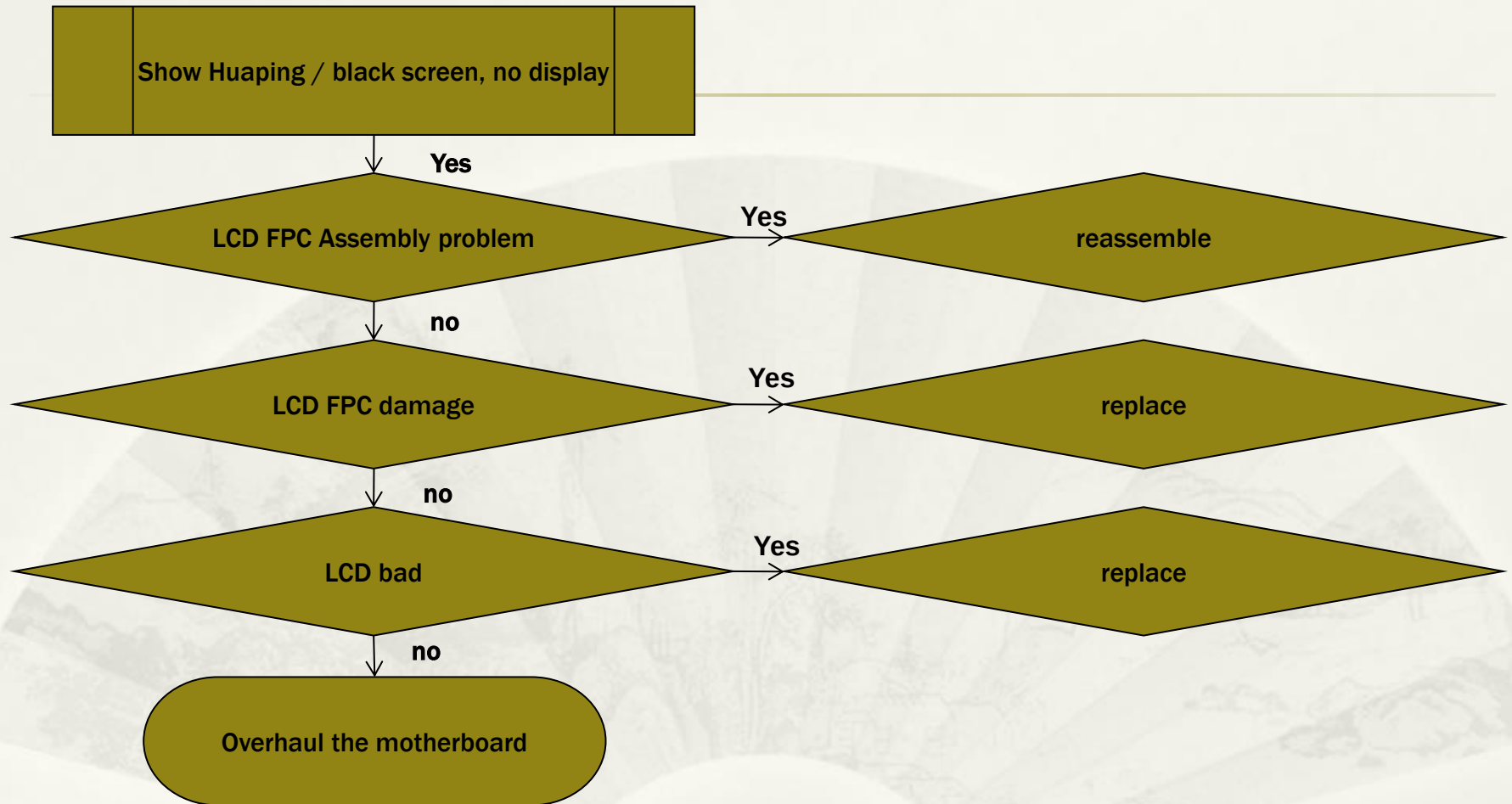
## 2.4 Can not boot failure analysis eMMC

--Confirm eMMC monomer U1801

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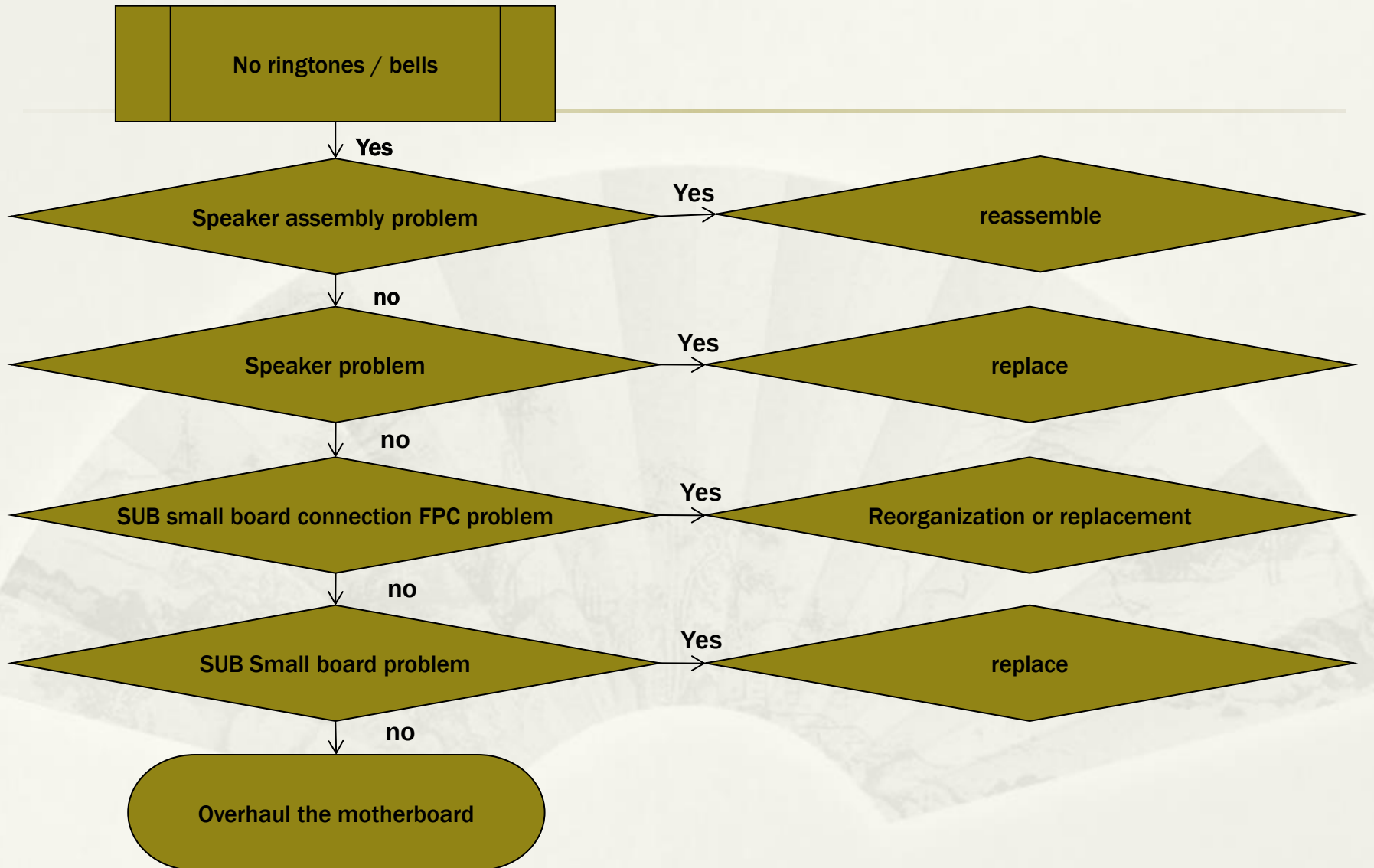
### 3. Display abnormal failure analysis



## 3.1 Display abnormal failure analysis LCM

- Make sure CON2301 is empty or missing ◦
- Has confirmed the completion of the boot, the measurement power supply VIO28\_PMU and VIO18\_PMU is normal
- Check whether the backlight is normal light, if the backlight does not need to check the backlight drive chip U0501 and its peripheral devices are normal. Backlight is the next step
- Measure whether all the signals on the R6108-R6115 are signaled,, and if not all signals, confirm that the CPU U1101 looks and X-ray, there is an abnormally solderable or further replacement

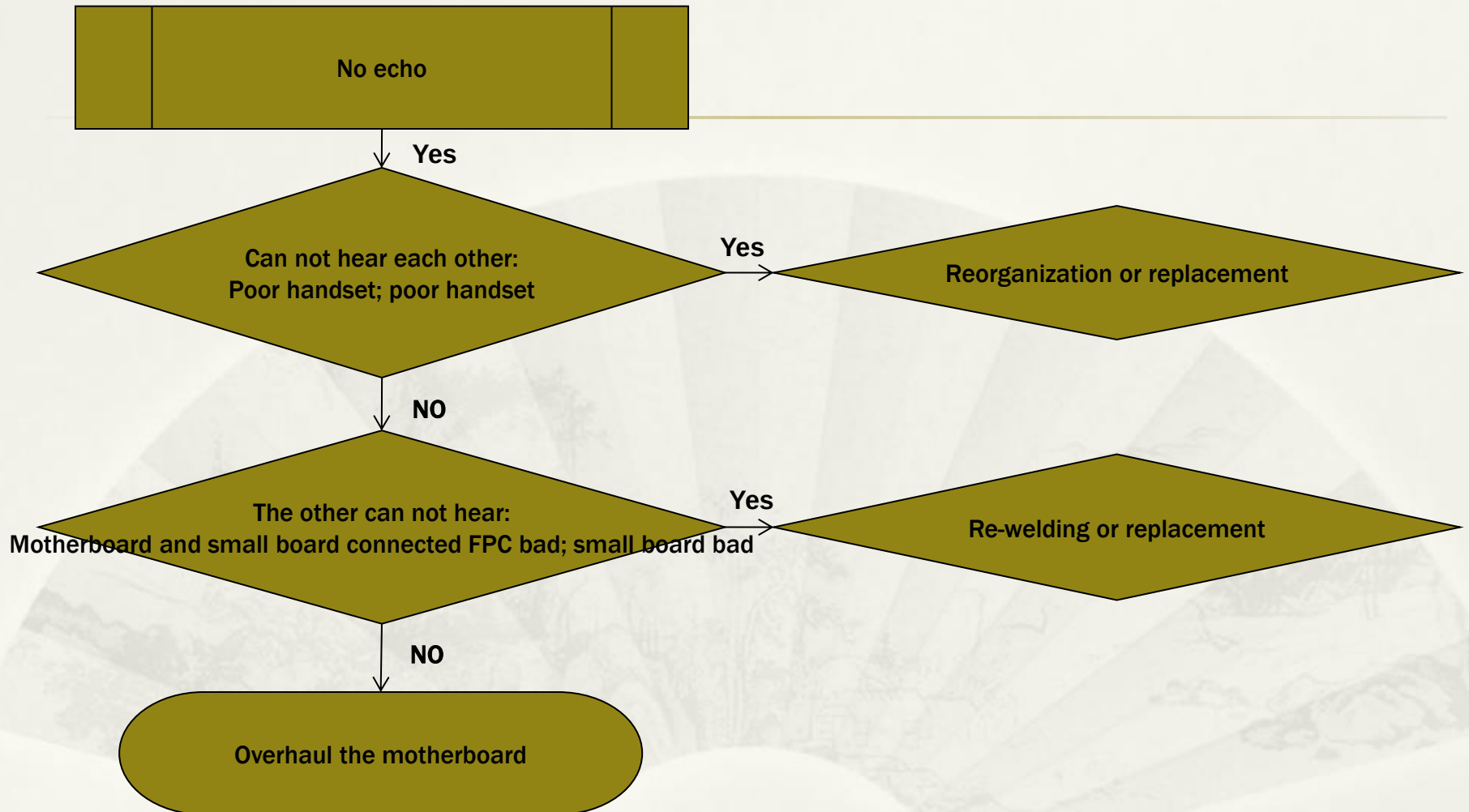
## 4. Ringtones fault analysis



## 4.1 Ringtones fault analysis

- Check B2109,B2110,C2111,C2112,C1001, C1003 are normal, abnormal re-welding or replacement.
- Make sure that the main PCB connector CON2701 has an empty weld or a loss item
- check PMIC U0701 appearance and X-ray, no exception can be welded or further replacement

## 5. No echo failure analysis

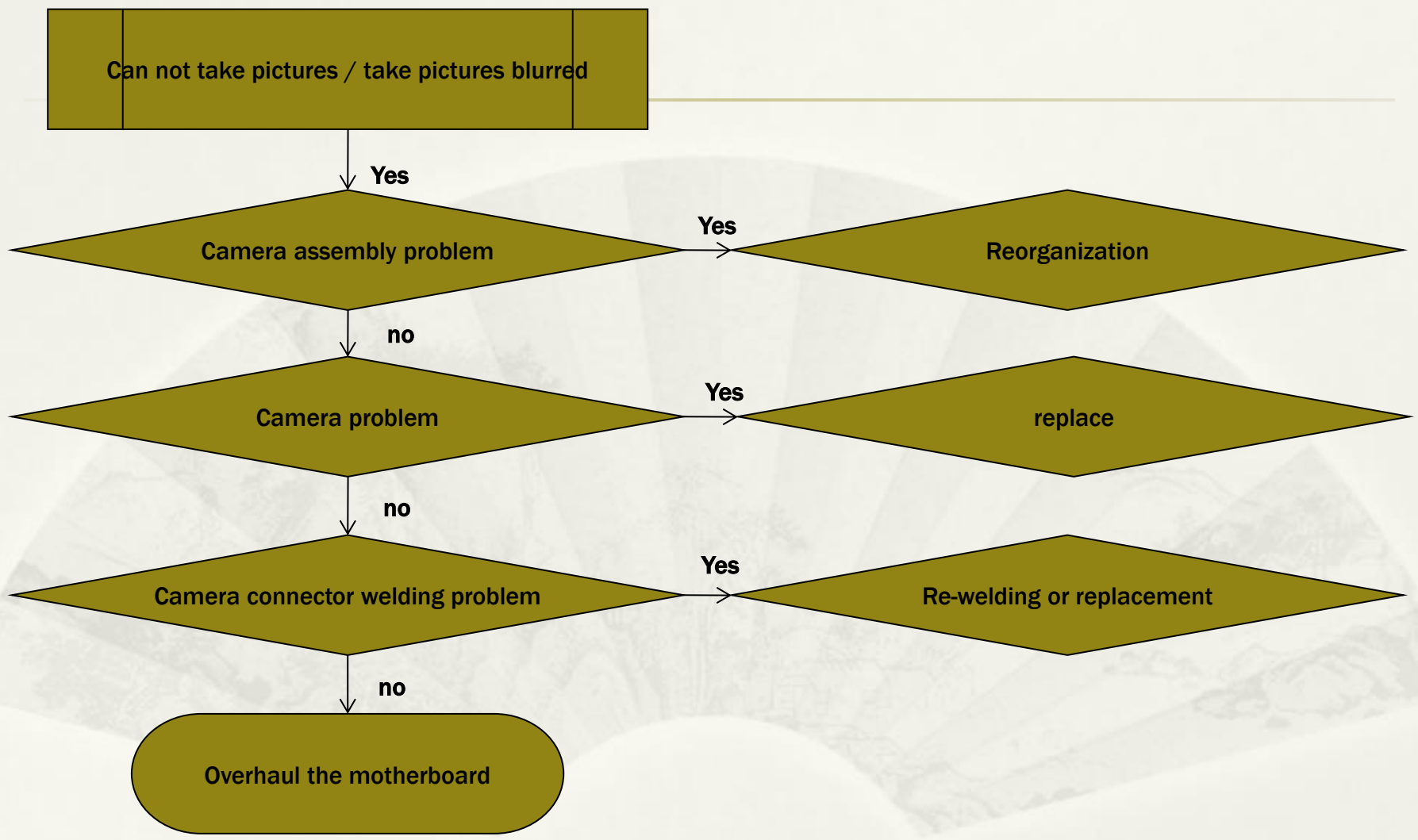




## 5.1 No echo failure analysis

- Check the peripheral material of the handset  
C2104,C2109,C2102,R2102, R2101,TVS2102,TVS2103  
whether there is any welding or missing parts 。
- Check sub board main MIC material  
C101,R1010,R1011,TVS102,C1007,C1013,C1014 have  
empty welding or missing parts 。
- Make sure that the main and subplane connectors CON1001  
pin17,18,20 CON2701 pin18,19,21 have Weld and the  
connector is damaged 。
- Confirm PMIC U0701 appearance and X-ray, no exception  
can be welded or further replacement

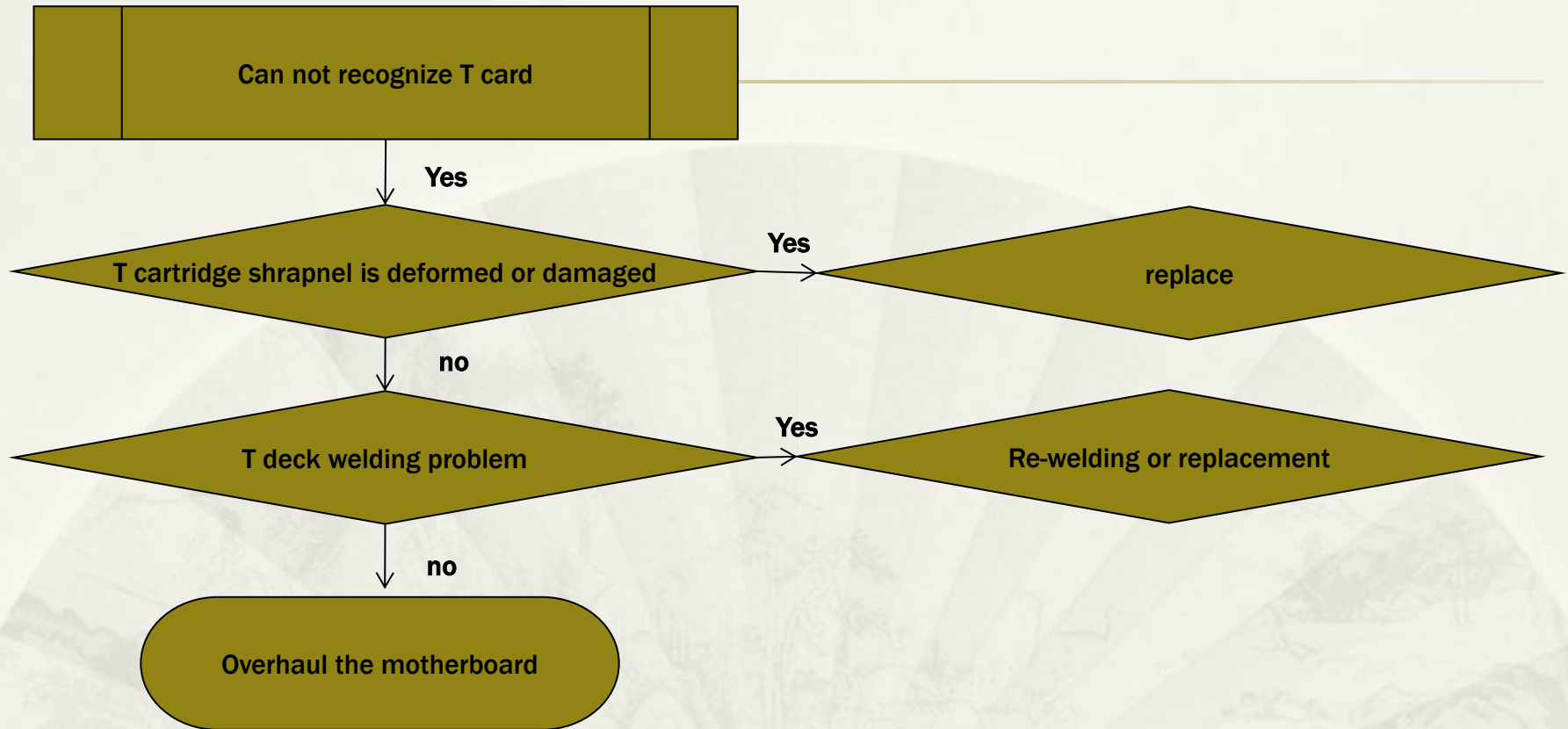
# 6. Poor camera failure analysis



## 6.1 Poor camera failure analysis

- Press the subject and the sub-picture to confirm whether the main connector CON2402, CON2401 and the sub-connector CON2403 are empty or missing
- Make sure the power supply is normal 。 。
- Check that the mains short circuit module R2431-R2440 is normal。  
Make sure that the sub-camera short circuit module R2415-R2442 is normal  
Make sure that the front-camera short circuit module R2443-R2452 is normal。
- Check camera switch U2401 is normal
- check U0701 and X-ray, no exception can be welded or further replacement

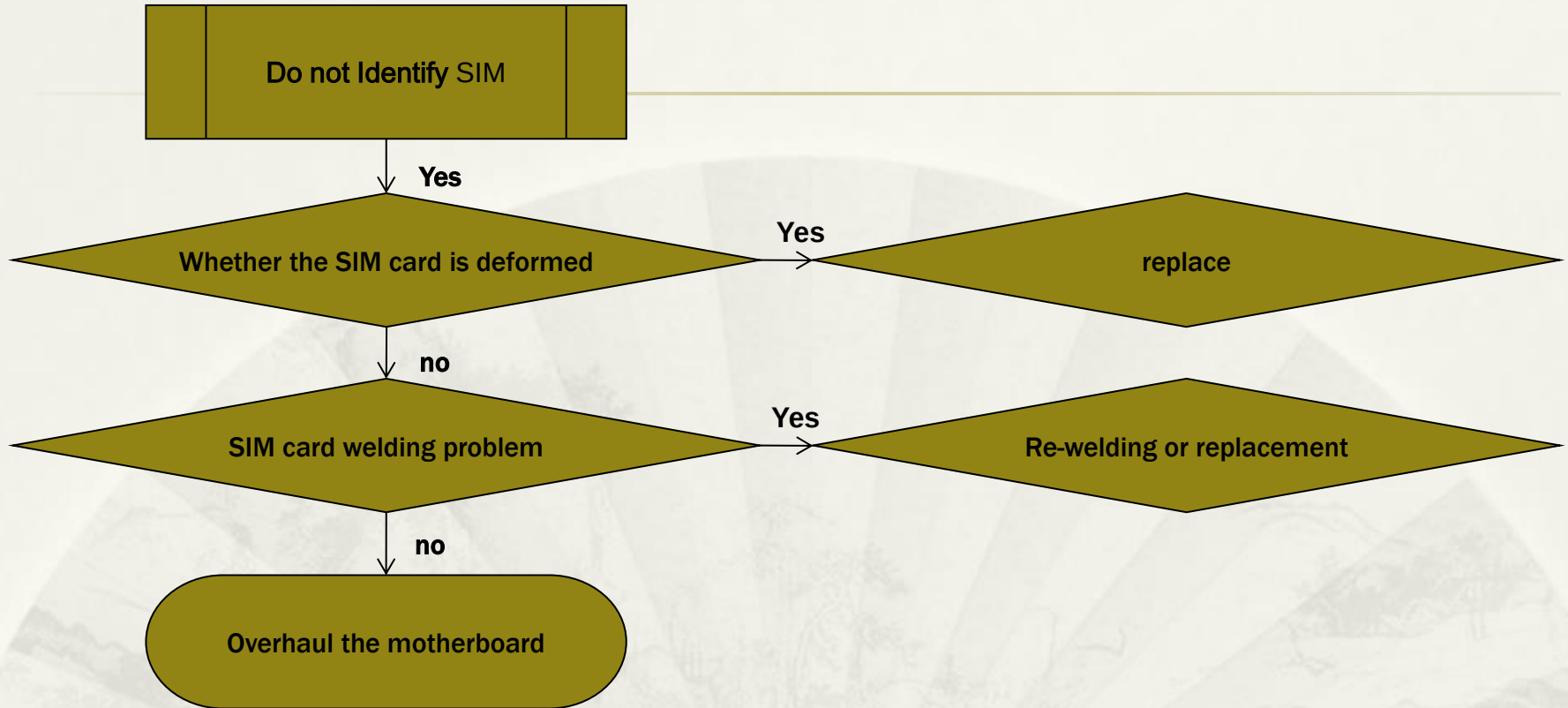
## 7.T card failure analysis



## 7.1 Fault Analysis of T Card

- Check if the relevant TVS tube has an empty weld or a loss item
- Check whether the power supply is normal, VMCH\_PMU output is 3.0V, confirm C2605,C2610。
- Check that the interrupt EINT\_SD is normal, CON2601 and peripheral devices 。
- Check CPU U1101 and X-ray, no exception can be welded or further replacement

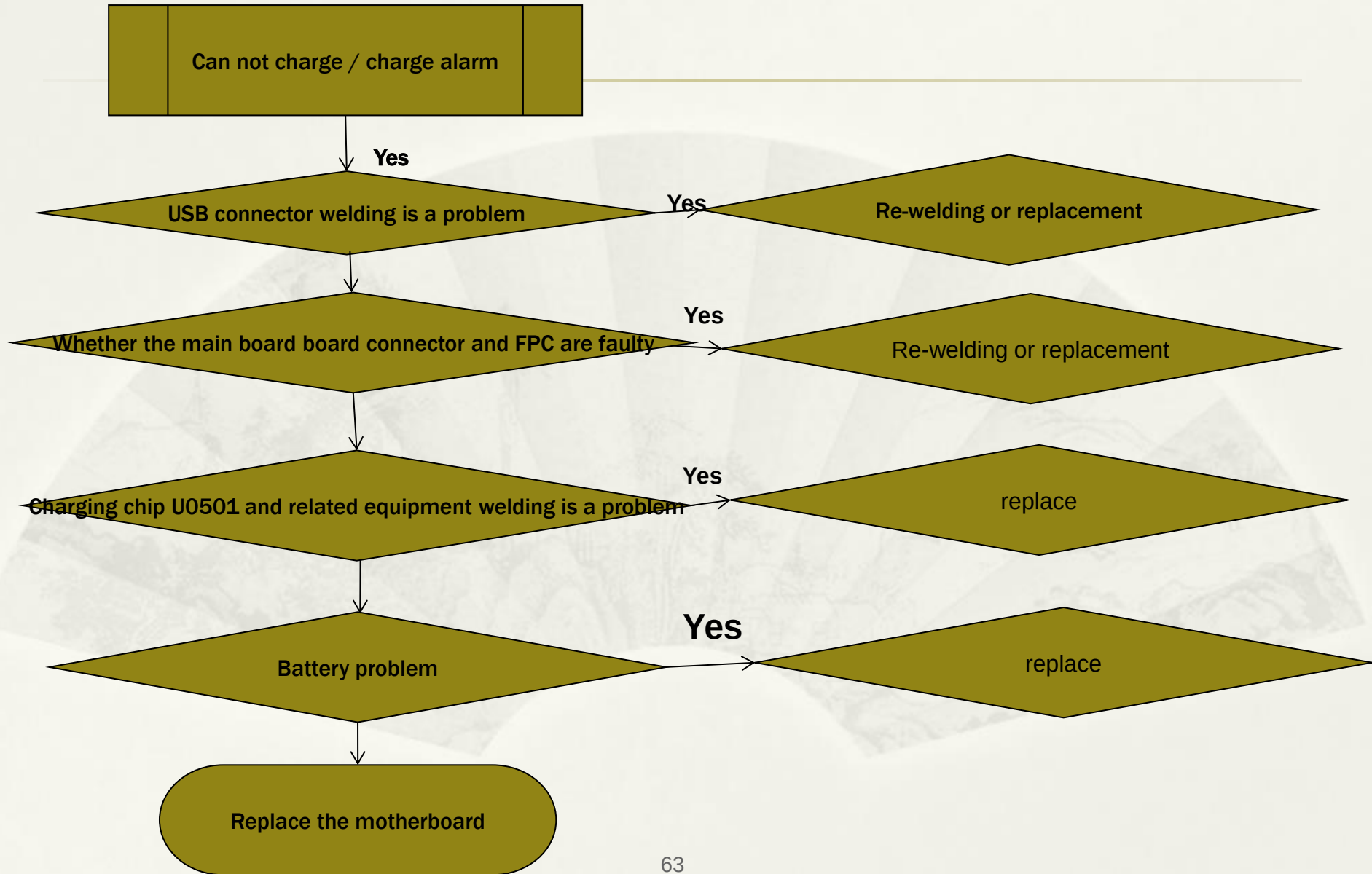
## 8. Do not Identify SIM failure analysis



## 8.1 Do not Identify SIM failure analysis

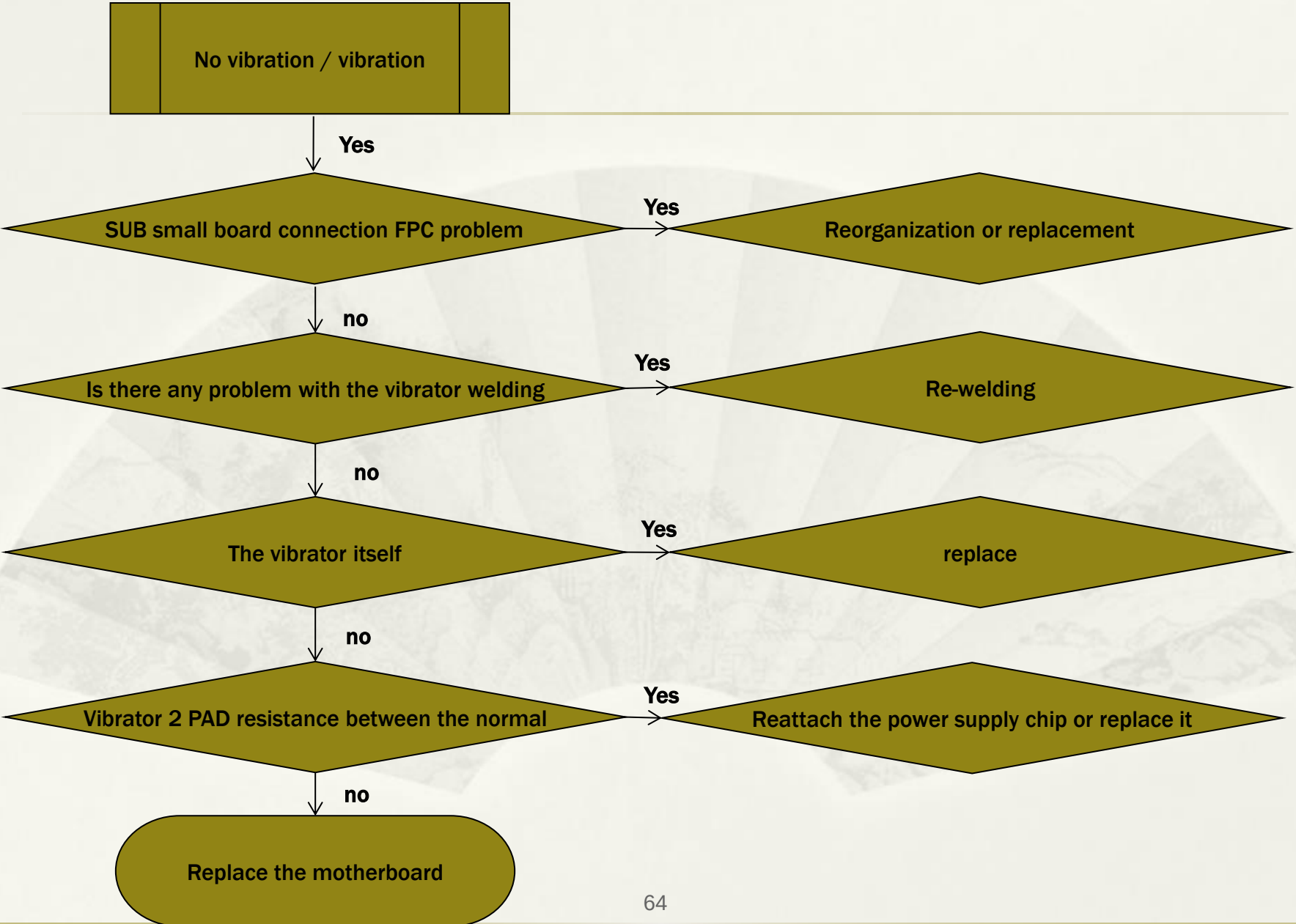
- Make sure that the SIM card connector CON2602,CON2603 is defective, empty, and damaged.
- To confirm whether the power supply is normal, VREG\_L14\_UIM1 、 VREG\_L15\_UIM2 output is normal 。
- Confirm interrupt UIM1\_PRESENT, UIM2\_PRESENT is normal.
- Confirm CPU U1001 and X-ray, no exception can be welded or further replacement.

## 9. Charge failure analysis

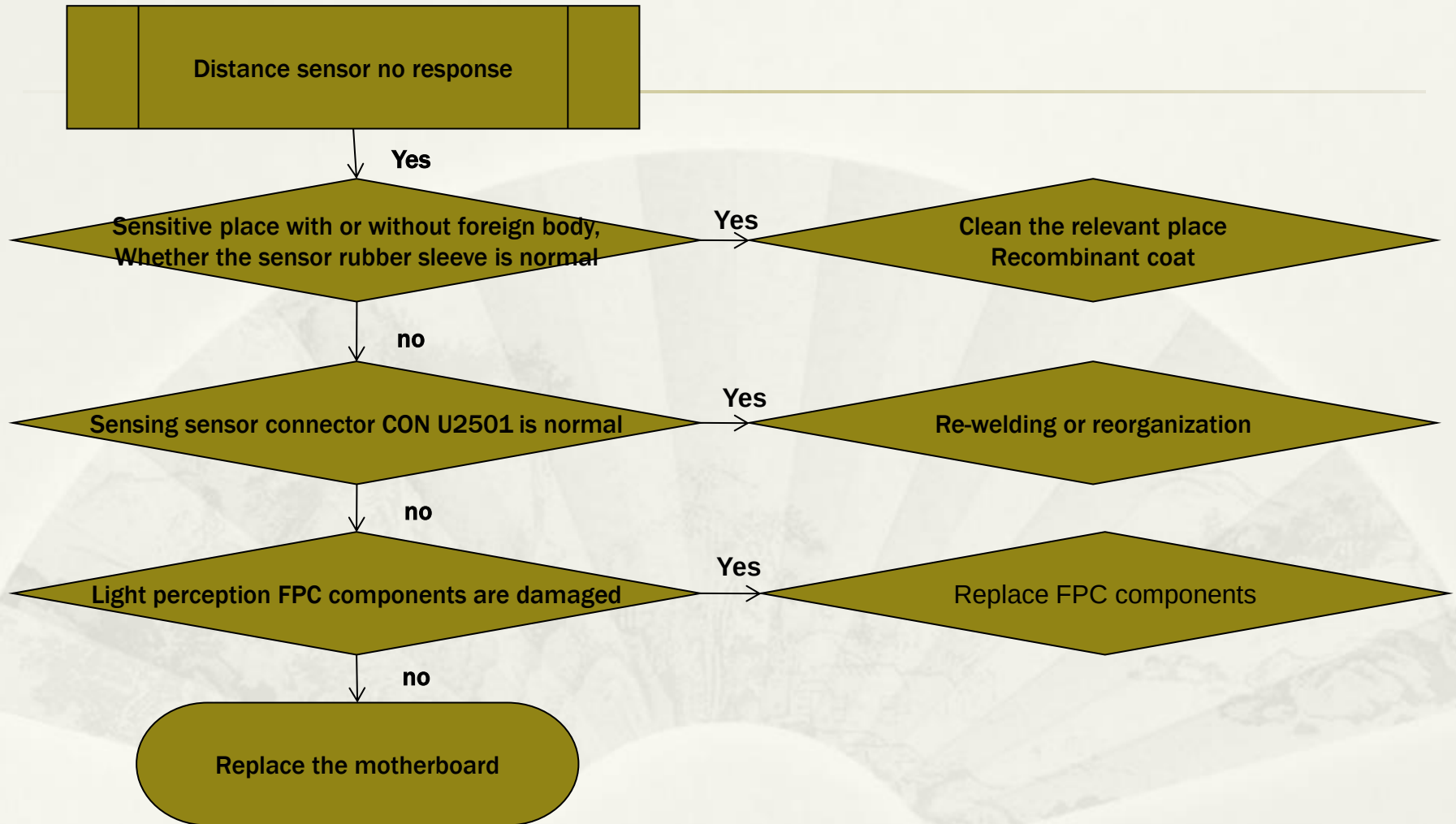




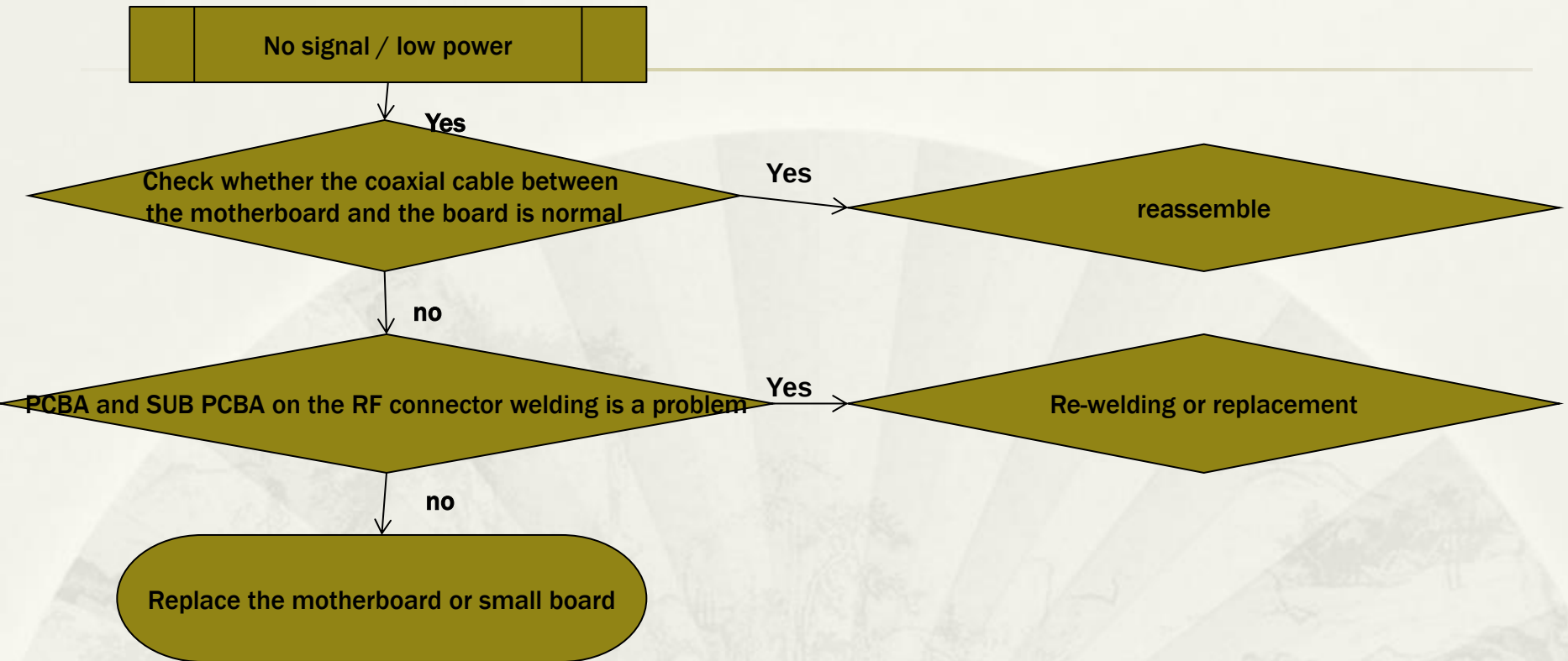
# 10. Vibration failure analysis



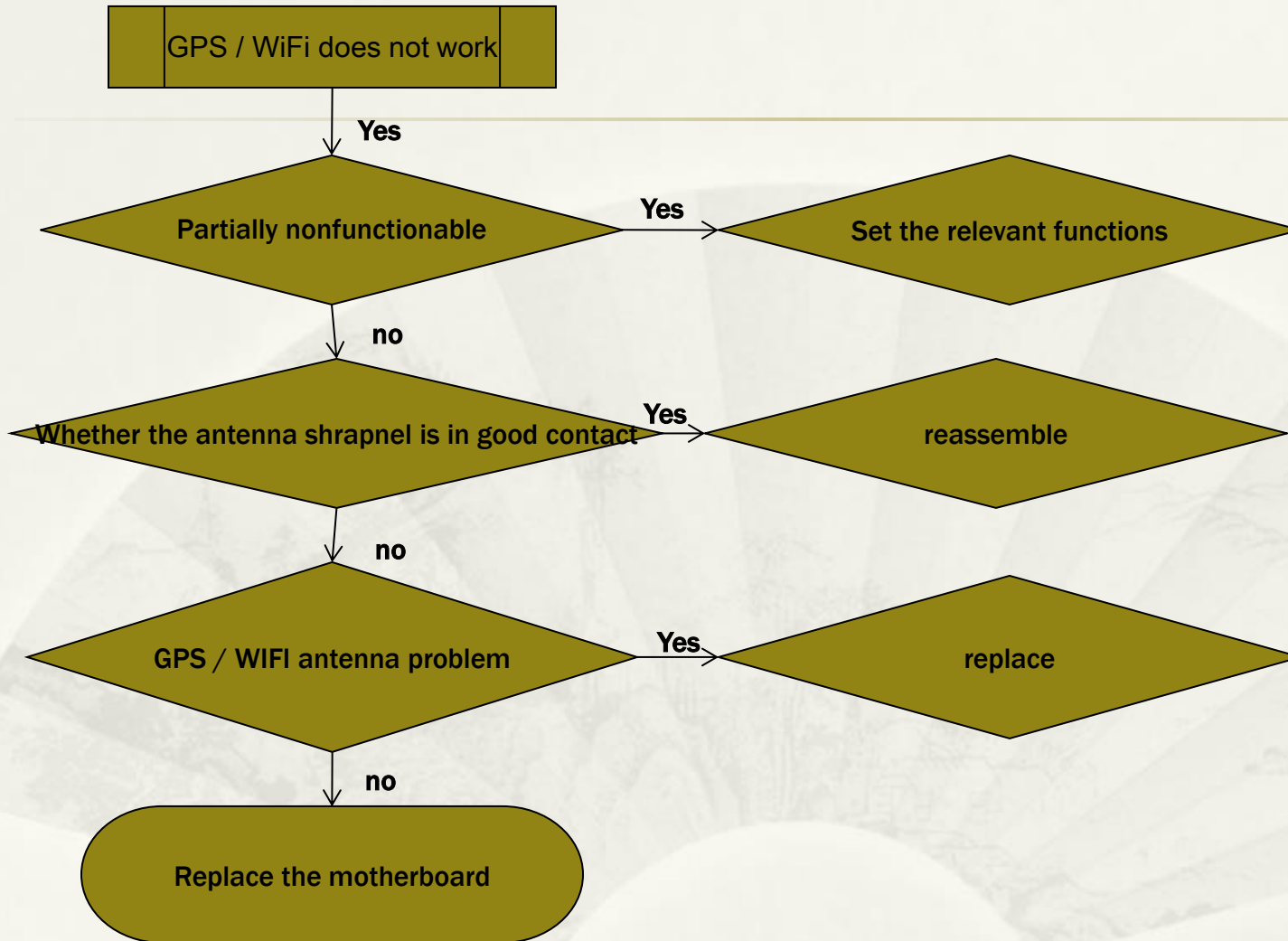
# 11. Analysis of Distance Induction Faults



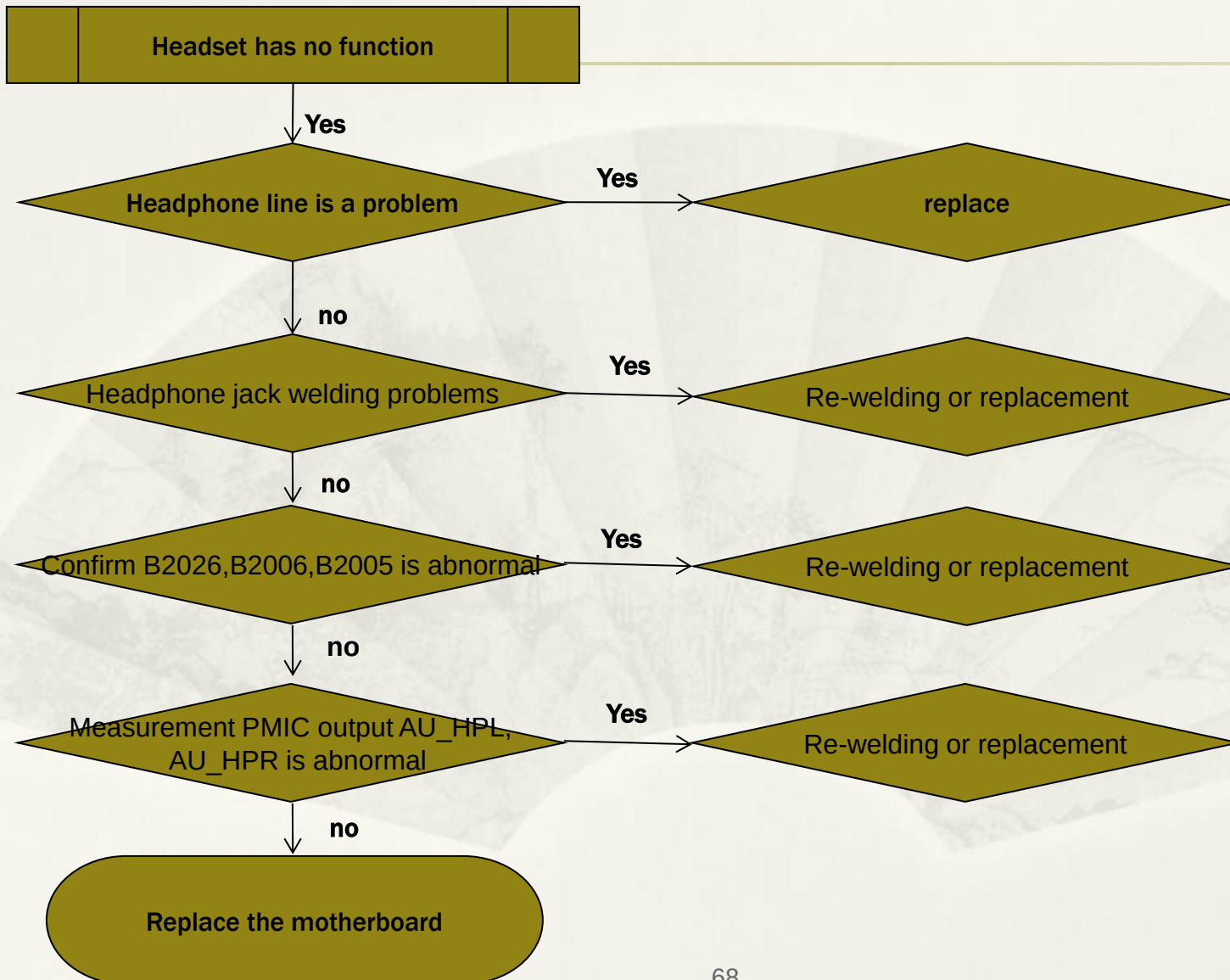
## 12. No signal failure analysis



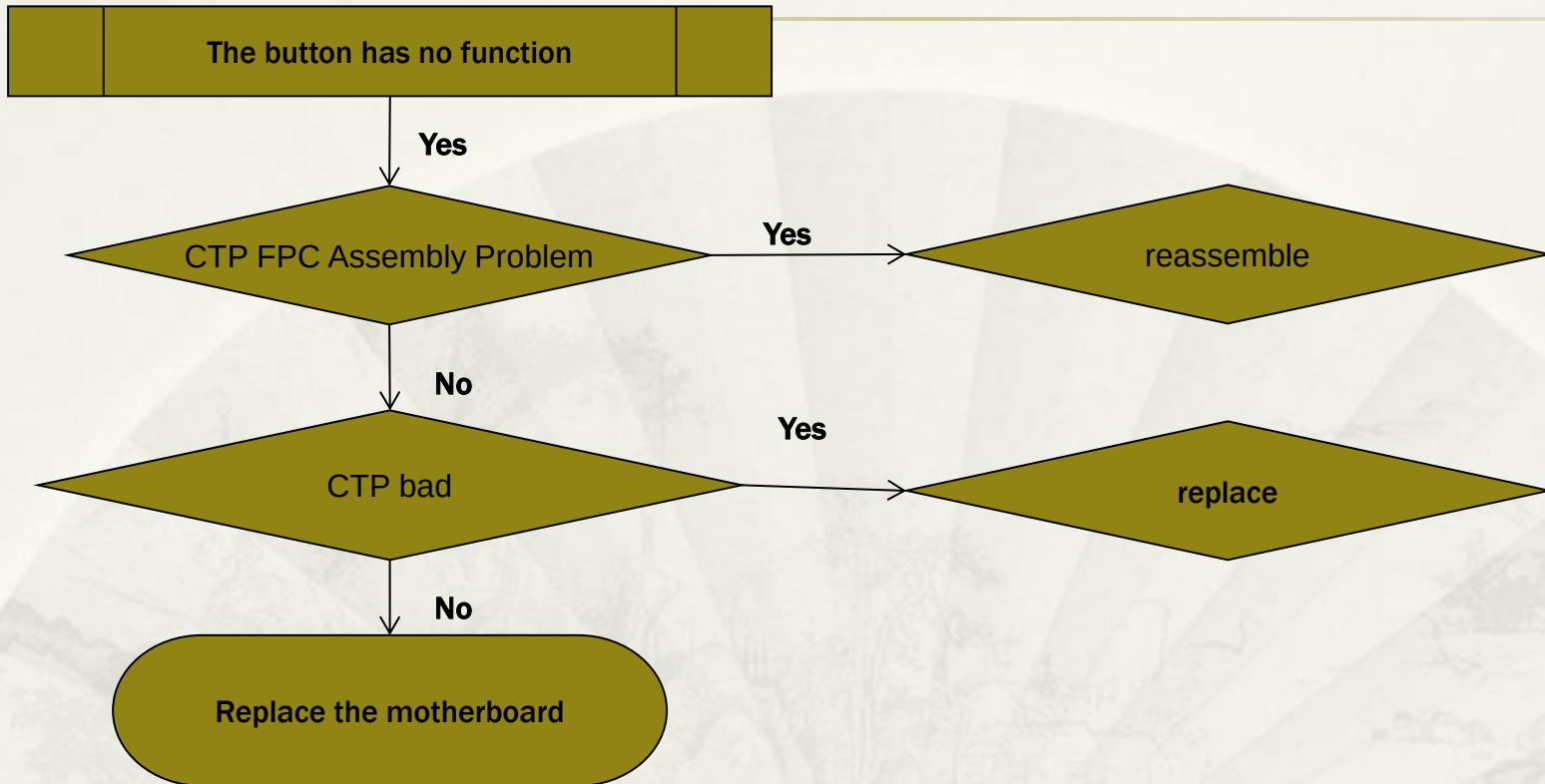
# 13. GPS / WiFi fault analysis



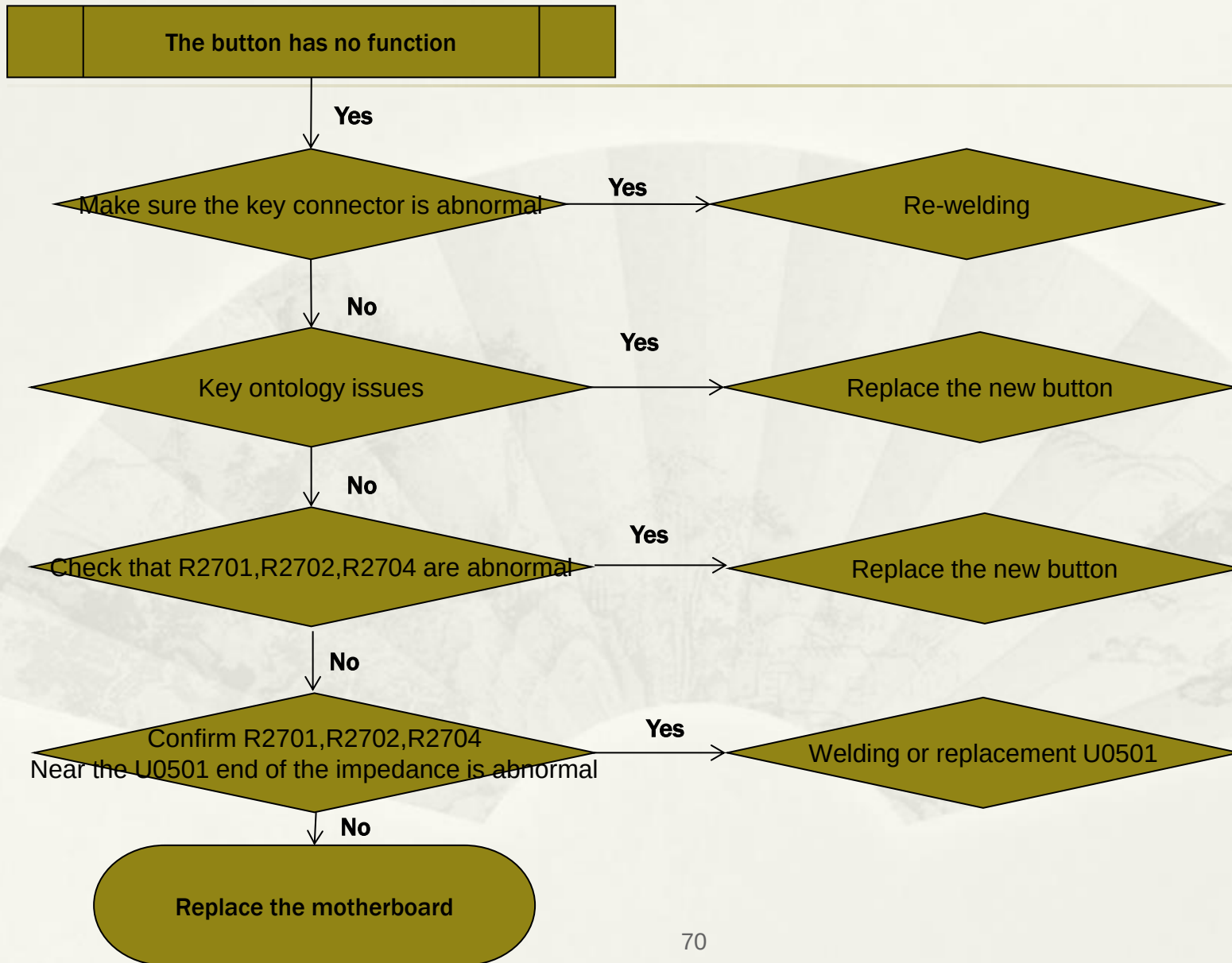
## 14. Headset problem analysis



## 15. Virtual Key or TP Failure Analysis



## 16. Key failure analysis





Thank you !